

PROJECT REPORT
ON
HEALTH MONITORING AND ALERT SYSTEM

In partial fulfillment of the requirements for the award of the degree of Bachelor of
Technology in Electronics & Communication Engineering
Maulana Abul Kalam Azad University of Technology

Submitted by

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ABSTRACT

Smart healthcare systems should be used by those who require ongoing monitoring, which is not available outside of hospitals. In rural or village areas, it is essential that nearby clinics may consult with city hospitals on the health of their patients. This model functions as a smart health monitoring system that checks a patient's condition using biomedical sensors and notifies those nearby using web technologies. Here, the biomedical sensors are linked to an ESP32 controller, which reads the data and interfaces it with a web server to display the results. This "Health Monitoring and Alert System Using IoT" goal is to develop a system capable of monitoring important body signals such as body temperature, heart rate, and SpO2 level are among examples. The sensors are synced with the controller and the web. A cloud server will receive the readings from the sensor sent by the device, and the data gathered will progressively be made available for analysis. The reading is saved on a cloud server, and the doctor's web portal or smartphone are also notified. These readings serve as both an analysis of the patient's health status and an emergency health condition alarm system. The controller, the web, and the sensors are all in sync. The device will transmit sensor readings to a cloud server, where the data will be gradually made available for study. It stores the reading on a cloud server and then sends an alert to the doctor's smartphone or web portal. These readings serve as both an analysis of the patient's health status and an emergency health condition alarm system.

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CHAPTER 1

INTRODUCTION

1.1 BACKGROUND

Hospitals are now heavily utilizing health care sensors. In recent years, researches have considered the use of medical sensors in clinical settings in remote health monitoring systems for long-term archiving, management, and clinical access to patient physiological data [1]-[4]. Patient checking monitoring is one of the main advantages of its inventive invention. A programmable remote health monitoring system is used to use an implanted device to measure the patient's body pulse and SpO2 level. The suggested system makes use of sensors such a pulse sensor, an oximeter and ECG sensor. These sensors are typically used to track sleeping habits, detect falls, and keep an eye on the patient's health.

Sensor units, coordinator units, remote server units, user interface units, and communication units are the main parts of health monitoring systems. A sensor unit consists of a number of battery-operated sensors for different health indicators, a computer for data processing, and an antenna for transmission [5]. To accomplish this, the system makes use of numerous sensors to keep track of vital indicators. These sensors can be connected to the doctor's smartphone or the internet. Sensor readings will be remotely transferred to the cloud via the gadget, and the resulting data will gradually become available for analysis. It has the ability to read and communicate emergency signs to the cloud, the doctor's smartphone, and eventually the doctor's website. These readings can be used to assess the patient's health and as an early warning system for a medical emergency.

1.2 STRUCTURE OF THE REPORT

The software used to perform simulation is Arduino IDE platform based on method of sensors. It analyses and controls different sensors simultaneously. It has been widely used in the design of prototype models ,embedded system , and other products. It can be used to take tracks of the data received from sensors and plot graphs and make changes according to it. This project shows a prototype model and analysis of the data with the help of WiFi module ESP32 micro-controller connected with discrete sensors. The project shows innovation of bio-medical platforms.

A total of 6 chapters make up the report. Beginning with the project's introduction, the second chapter explains the ESP32 microcontroller, its applications, advantages, and limitations. In the third chapter the sensors like LM35,Pulse Oximeter and AD8232 ECG. Sensors assembled with help of jumper wires and simulated using Arduino IDE software. Due to compact model these will be further modified and experimented in the coming future. In the fourth chapter, the model assembling , programming is done of discrete sensors with controller. In the fifth chapter, working of the prototype model is designed and simulated. Working on IOT sensors are described in this chapter. In the sixth chapter, impact assessment, risk issues, cost analysis are described.

CHAPTER 2

THEORY OF PROTOTYPE MODEL

This chapter provides background information regarding ESP32 controller. The structure of ESP32 module along with its advantages, disadvantages.

2.1 ESP32 Micro-controller

The ESP32 is a low-cost, low power micro-controller with built-in Bluetooth and Wi-Fi.

It is an upgraded version a low-cost Wi-Fi microprocessor with a few features.

The power amplifier, filters, antenna, RF balun, low-noise amplifiers, and power management module are all combined into one unit. A very little part of the printed circuit board is used for the entire area. With the best RF and power characteristics, as well as being safe, dependable, and extensible to a variety of applications, this module employs TSMC 40nm low power 2.4Ghz dual-mode Wi-Fi and Bluetooth chips.

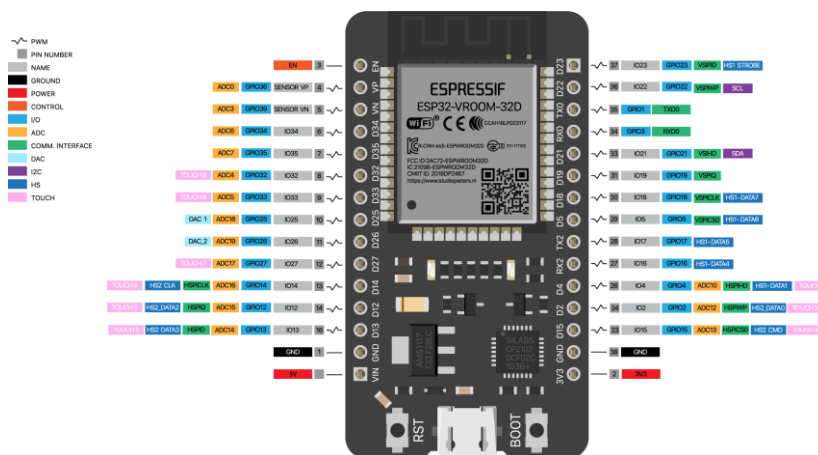


Fig: 2.1 ESP32 Module

The ESP32 Specifications-

1. Dual-Core 32-bit LX6 microprocessor ESP WROOM32
2. SRAM: 520KB, ROM: 448KB, and support up to: 16MB
3. CP21XX USB-to-UART(serial) Bridge built-in
4. 802.11b/g/n/e/I Wi-Fi
5. Bluetooth version 4.2 with BR/EDR and BLE.
6. 2 x 8-bit DACs
7. 9 x touch sensors total Note: [D0] represents touch sensor 1. However, this specific ES32 module does not have it as a pin (version with 30 GPIOs). The 36-pin variant has access to the GPIO 0 pin.
8. Internal Temperature Sensor

Pros of ESP32 Module

1. Both development boards and independent chips are provided.
2. When each of the 36 I/O pins can be configured to perform various purposes, multiplexing.
3. Built in Wi-Fi module and Hall effect sensor.
4. Bluetooth 4.2 and Bluetooth low energy support.
5. Contacts that can be touched to jolt the micro-controller out of a deep sleep.
6. There are many resources available for developing firmware, including the Arduino IDE environment, Micro-Python, an environment for the ESP Home that does not require programming and is reasonably priced.

Cons of ESP32 Module

1. They won't connect to a contemporary 5 GHz network; instead, they only support 2.4 GHz Wi-Fi networks.
2. Require more GPIO pins for working with extra sensors.

CHAPTER 3

SENSORS

In this chapter MAX30100 pulse oximeter and I2C Interfaced 16X2 LCD background information will be provided along with specification.

3.1 MAX30100 Pulse Oximeter Sensor

To function, every organ system in the body needs oxygen. Cells start to malfunction and eventually perish when oxygen is not present. It may result in serious symptoms and, in the end, organ failure.

The oxygen is filtered by the lungs before being supplied to the organs. Following that, it is transferred into the blood by proteins in red blood cells known as haemoglobin, and further proteins give oxygen to the rest of the body [6].

The oxygen saturation level in haemoglobin proteins is measured by pulse oximeter. Saturation of oxygen is determined by how much oxygen is reaching the organs.



Fig: 3.1 Pulse Oximeter

The range of normal oxygen saturation is from 95 to 100 percent. Oxygen saturation levels below 90% are regarded as excessively low and may indicate a clinical emergency.

3.2 Overview of the MAX30100 Module Hardware

The module includes an Analog Devices MAX30100 Pulse Sensor, a contemporary combined IC of pulse oximeter and heart sensor which uses two LED's, an improved optics system, a photo detector, and low-noise analogue signal processing to detect heart rate (HR) and pulse oximeter (SpO2) signals. The MAX30100 includes two LED's -a red and infrared LED-on the right and a highly sensitive photo detector on the left. According to the concept, blood oxygen levels and heart rates can be determined by emitting a single LED at a time, measuring the amount light that bounces back at the detector, and interpreting the signature [7].

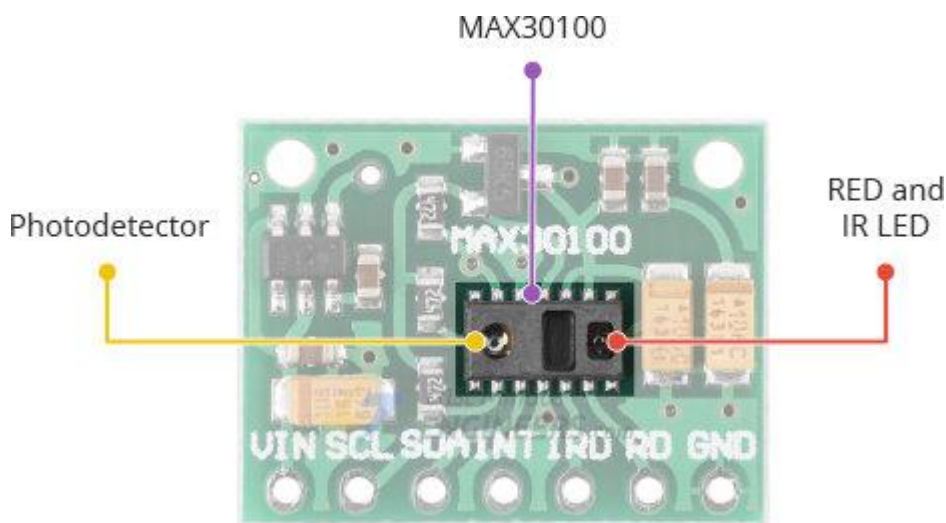


Fig: 3.2 Pulse Oximeter (Photo Detector)

Two supply voltages are needed by the MAX30100 chip: 1.8V for the IC and 3.3V for the RED and IR LED's. So, the 3.3V and 1.8V regulators are included with the Pulse oximeter module. This enables you to connect the module to any micro-controller that has I/O at levels of 1.8V, 3.3V, or even 5V.

The Pulse oximeter's (MAX30100) low power consumption, which is less than 600uA during measurement, is one of its most significant advantages. When the MAX30100 is operating in standby mode, it uses just 0.7uA. Implementation in battery-operated devices, such as phones, wearable, or smart watches, is simple due to minimal power consumption.

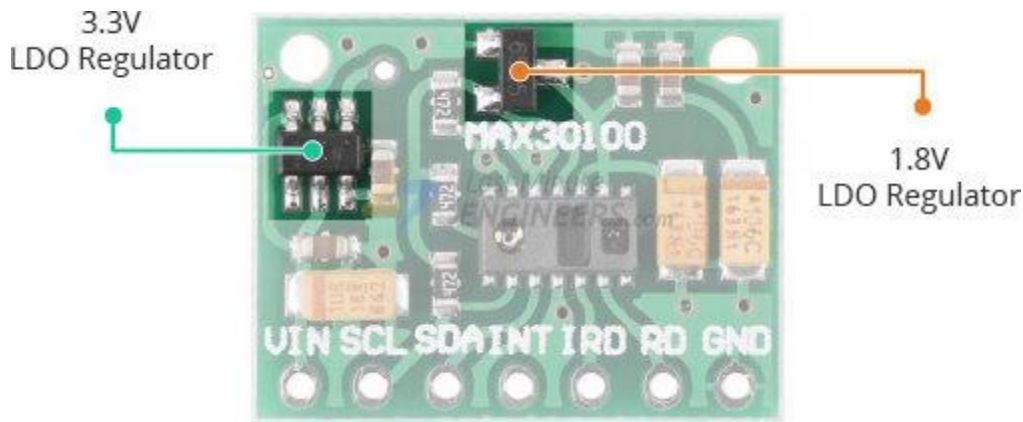


Fig: 3.2.1 Pulse Oximeter(Regulator)

3.3 MAX30100 Working

A pair of high-intensity LEDs (IR and RED, which have different wavelengths) and a photodetector are the main components of any optical pulse oximeter and cardiac sensor, including the MAX30100. The wavelengths of these LEDs are 880 nm and 660 nm, respectively.

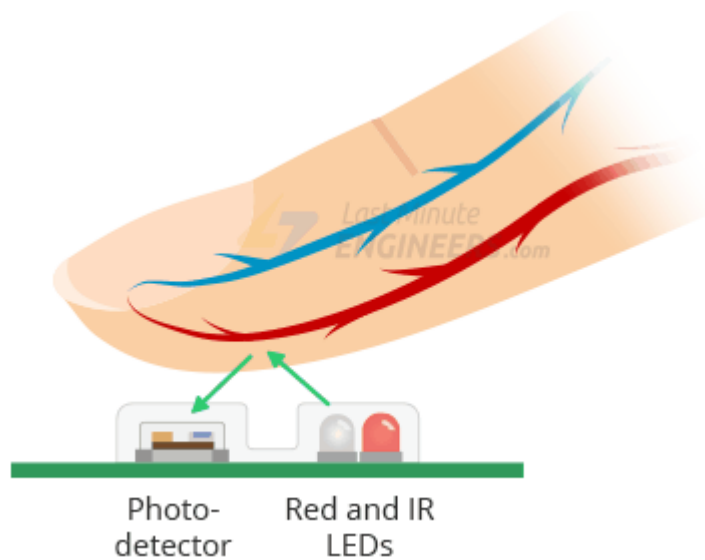


Fig: 3.3 MAX30100 Working

Once the MAX30100 emits both lights onto the finger or earlobe (or really anywhere the skin isn't too thick, allowing both lights to readily penetrate the tissue), a photo-detector is used to quantify the amount of reflected light.

The two elements that make up the MAX30100's operation are the pulse oximeter, which measures the blood's oxygen content, and heart rate measurement.

Heart Rate Sensor

The arterial blood contains oxygenated haemoglobin (HbO_2), which has the ability to absorb IR radiation. The more the haemoglobin, which is shown by redder blood, the more IR radiation is absorbed. The amount of reflected light, which results in a waveform at the photo-detector's output, is caused by the blood pumping through the tips of fingers [9]. We rapidly begin to acquire of the heartbeat (HR) pulse thanks to the light generated by the LED and reflected by the tip of the finger.

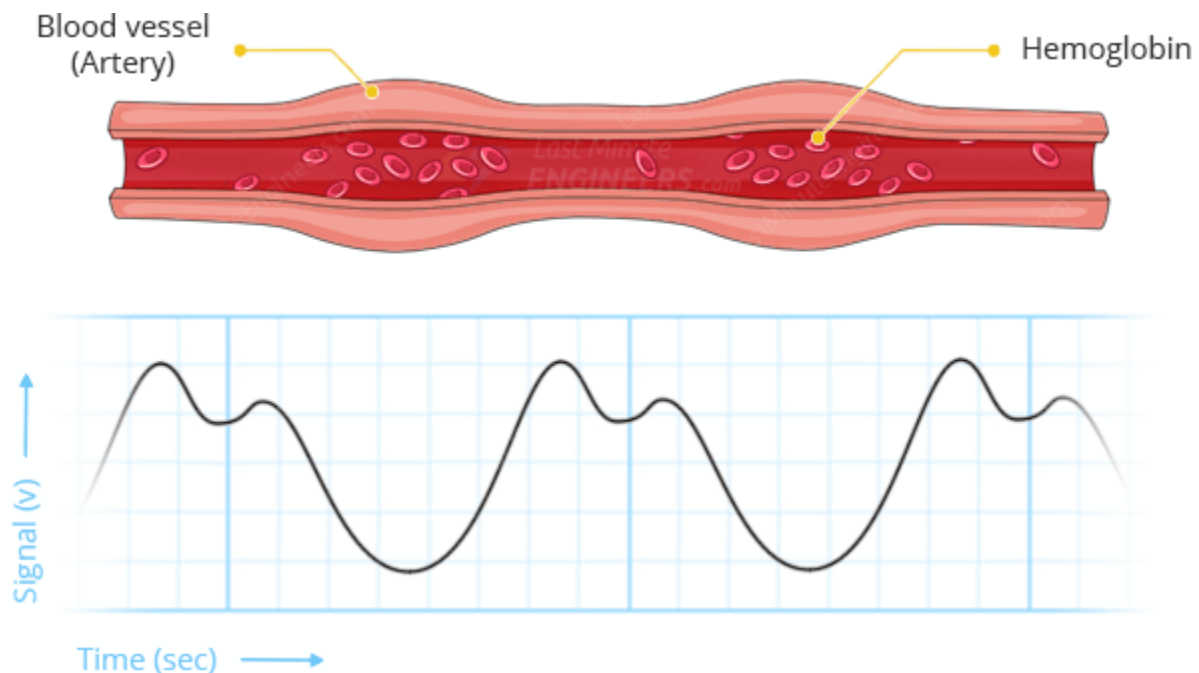


Fig: 3.3.1 Blood Vessel

Pulse Oximetry

Working on the theory that the amount of RED and IR light absorbed varies with blood oxygen levels. The absorbance spectra of oxygenated and deoxygenated haemoglobin is shown in the graph below (Hb).

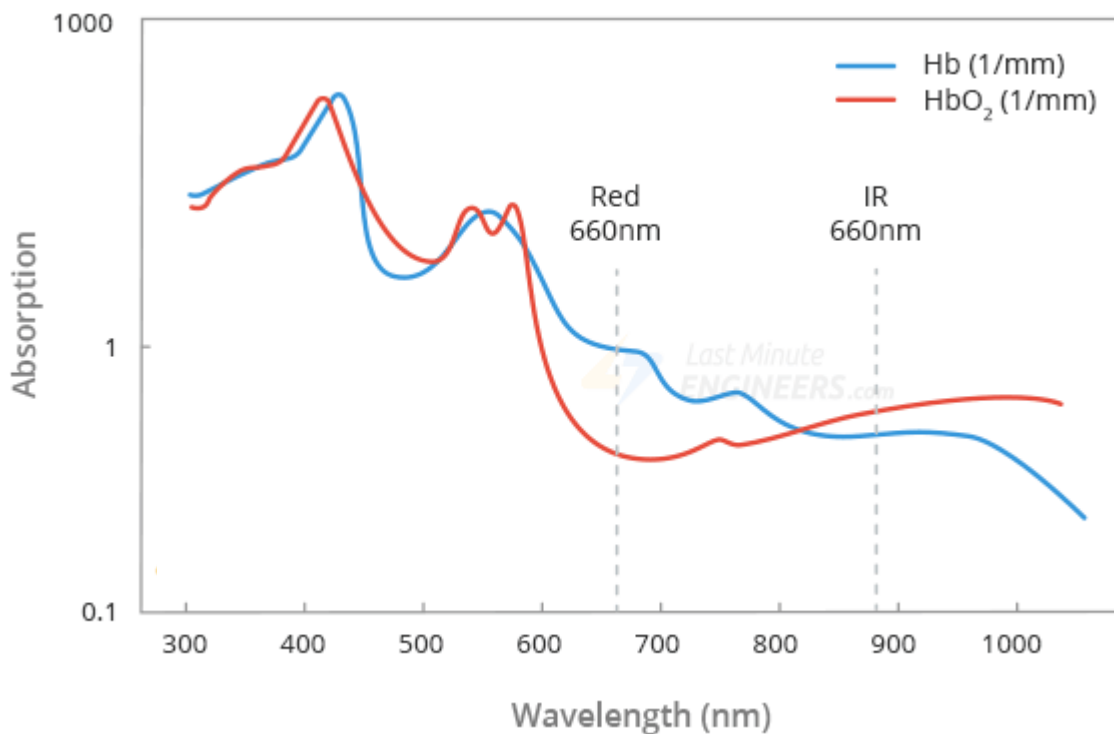


Fig: 3.3.2 Absorption vs Wavelength

Deoxygenated blood, as depicted in the graph, absorbs more red light (660nm), but oxygenated blood absorbs more infrared light (880nm). The oxygen content (SpO₂) of the blood is determined after assessing the photo-detector's sensitivity to IR and RED light.

3.4 Pin-out for MAX30100 Module

The connections below are produced by the MAX30100 module.

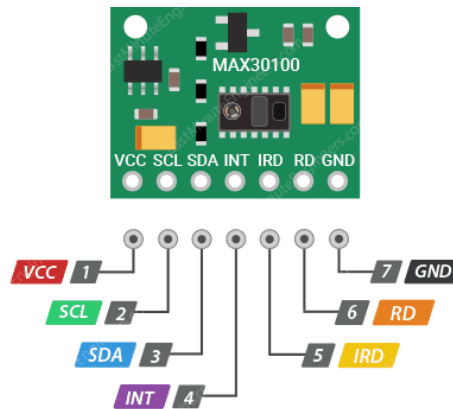


Fig: 3.4 MAX30100 Pin-out

VIN The power pin is VIN. It can be connected to your ESP32 module's 3.3V or 5V output.

SCL Connect your ESP32's I2C clock line to the SCL pin, which is the I2C clock pin.

SDA Connect your ESP32's I2C data line to the SDA pin, which is the I2C data pin.

INT For each pulse, the pin can be programmed to produce an interrupt. Since this line has an open drain, the internal resistor pulls it HIGH. The pin becomes low when an interrupt happens and remains low until the interrupt is cleared.

IRD In order to drive LED pulses for SpO2 and Pulse Rate(HR) measurements, the MAX30100 includes an LED driver. If you don't want to run the IR LED, leave it unplugged.

RD The IRD pin and the RD pin are quite similar, but the RD pin is used to drive the RED LED. If you don't want to drive the RED LED, leave it unplugged.

GND The ground is GND.

3.5 I2C 16X2 LCD

An I2C 16x2 LCD screen with an I2C interface is seen here. It can show 16 characters by 2 rows, each with a white character on a blue background. I2C 16x2 LCD is favoured since projects involving Arduino LCD demand more pins and are difficult to connect and solder. I2C is the communication protocol used by this 16x2 Arduino LCD screen. It means that VCC, GND, SDA, and SCL are the only pins required for the LCD display. It will conserve at least 4 ESP32 digital or analogue pins. Every connector is an XH2.54 standards (Breadboard type). You can immediately connect to the jumper wire.

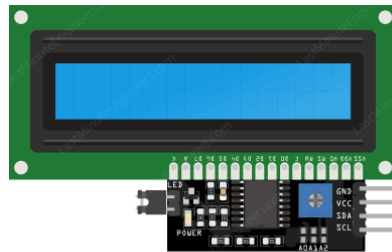


Fig: 3.5 I2C 16X2 LCD

3.6 Pin-out for I2C LCD Display

Only 4 pins connect an I2C 16x2 LCD to the outside world. Following are the connections:

GND stands for ground pin. Connect it to the ESP32 module's ground.

VCC The module and LCD are powered by VCC. Connect it to a 5V external power supply or the ESP32's output.

SDA The I2C data pin is SDA. Connect it to the I2C data pin on the ESP32 module.

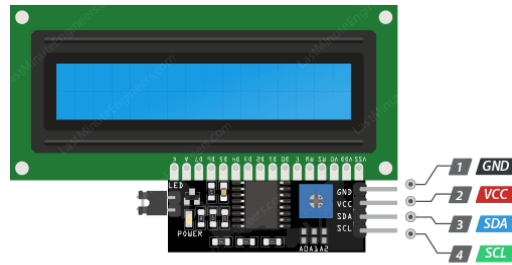


Fig: 3.6 I2C Pin-out

3.7 Hardware Overview I2C LCD

An I2C LCD adapter and a character LCD display powered by the HD44780 comprise the I2C 16x2 LCD display.

Character LCD Display

These LCD's are perfect for showing solely text or characters, as the name implies.

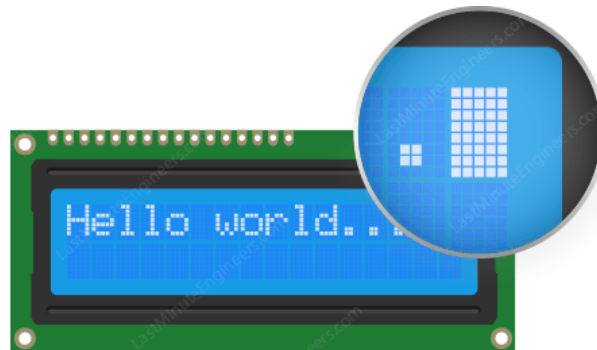


Fig: 3.7 I2C LCD

For instance, a 16x2 I2C LCD can show up to 32 ASCII letters in two rows of 16 each
And has an LED backlight.

I2C LCD Adapter

The core of the adapter is the expanded chip, the PCF8574 8-bit I/O. This chip transforms the I2C data from the ESP32 into the parallel data needed for an LCD display.

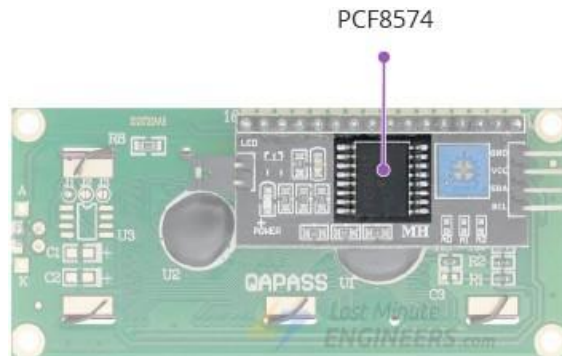


Fig: 3.7.1 LCD CHIP

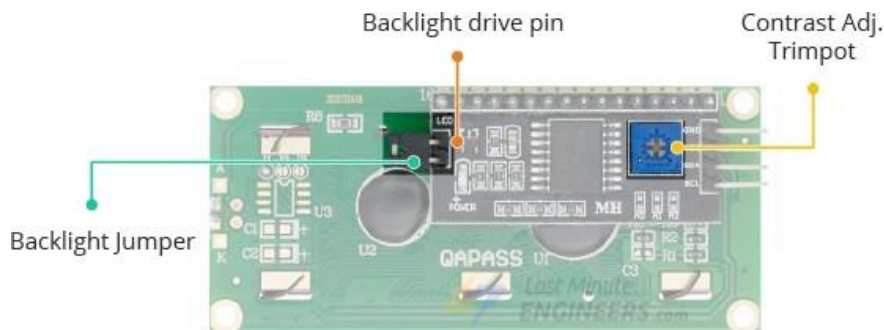


Fig: 3.7.2 LCD CHIP

A tiny Contrast Adj.trim-pot is also included on the board to allow for precise contrast alterations. The jumper pin on the PCB that powers the backlight is also there. You can take off the jumper pin and use an external voltage source to adjust the back-light's brightness by connecting it to the header pin with the designation “LED”.

3.8 AD8232 ECG SENSOR

The electrocardiography, also known as the ECG, is a method for collecting electrical impulses produced by the human heart. The ECG sensor allows us to identify the degree of physiological arousal when a person experiences it, but it is also used to comprehend a person's psychological state. In order to determine the electrical action of the heart, an AD8232 sensor is employed. This is a tiny chip, and its electrical behavior can be visualized using an ECG-like graphic.

The AD8232 ECG monitor is a commercial board that measures the electrical activity of the human heart. This activity yields an analogue reading that can be shown on a chart that resembles an ECG. Electrocardiograms can be highly loud, so the AD8232 chip can be used to reduce the noise. To help extract a clear signal from the intervals, the ECG sensor performs similarly to an operational amplifier. The AD8232 sensor is utilised in ECG signal conditioning and biopotential measuring applications. The main purpose of this chip is to improve, retrieve, and filter weak bio-potential signals in noisy situations, such as those produced by removing a remote electrode and motion.

3.9 AD8232 Pin Configuration

The heart rate tracking sensor, like the AD8232, has the pins SDN pin, LO+ pin, LO-pin, OUTPUT pin, 3.3V pin, and GND pin. So that we may connect this IC to development boards like Arduino by attaching pins to them. The right arm (RA) and left arm (RL) pins, among other custom sensors, have connectors on this board. The human pulse rhythm is displayed on this board using an LED indication.

The AD8232 sensor has a fast restore feature that is used to shorten the HPFs' lengthy resolving tails. This sensor is available in a 4mm x 4mm dimension, and it comes in a

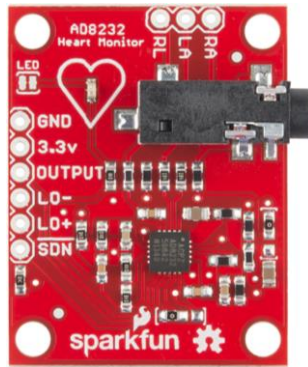


Fig: 3.9 AD8232 ECG

20-lead LFCSP package. Although it can work between -40 and +85 degree Celsius, its performance range is 0 to 70 degrees.

3.10 AD8232 Applications

The following are some examples of the AD8232 ECG sensor's applications:

- Monitoring heart rate and physical activity
- Simple ECG
- Capturing a bio-potential signal
- Biomedical instrument prototyping
- Human-computer interaction
- Fluctuation in heart rate
- Bio-metrics

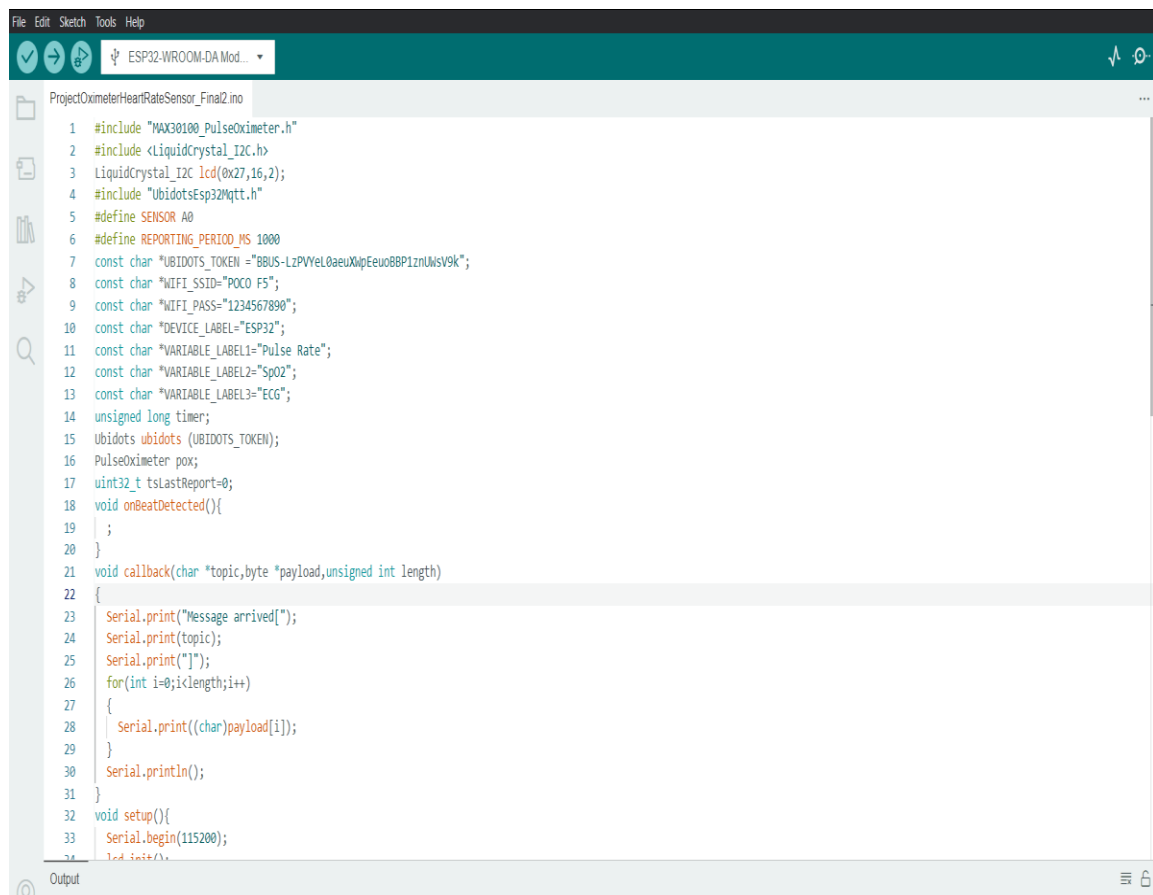
As a result, the AD8232 ECG sensor is a sensitive, tiny chip used to monitor the electrical activity of the heart and measure it (ECG).

CHAPTER 4

PROGRAMMING ESP32 MODULE

In this chapter we are working on the coding part of the ESP32 module interfaced with sensor MAX30100 Pulse Oximeter Sensor, AD8232 ECG and I2C Interfaced 16X2 LCD using jumper wires. The libraries required to download before writing the code are PubSubClient ,Ubidots ESP MQTT and LiquidCrystal I2C. We have used Arduino support programming language coded in Arduino IDE platform an easy IDE platform for simulation and working.

Coding ESP32 Module



```
1 #include "MAX30100_PulseOximeter.h"
2 #include <LiquidCrystal_I2C.h>
3 LiquidCrystal_I2C lcd(0x27,16,2);
4 #include "UbidotsEsp32Mqtt.h"
5 #define SENSOR_A0
6 #define REPORTING_PERIOD_MS 1000
7 const char *UBIDOTS_TOKEN = "8BUS-LzPVVeL0aeuXhpEeu0BBP1znUmsV9k";
8 const char *WIFI_SSID="POCO F5";
9 const char *WIFI_PASS="1234567890";
10 const char *DEVICE_LABEL="ESP32";
11 const char *VARIABLE_LABEL1="Pulse Rate";
12 const char *VARIABLE_LABEL2="SpO2";
13 const char *VARIABLE_LABEL3="ECG";
14 unsigned long timer;
15 Ubidots ubidots (UBIDOTS_TOKEN);
16 PulseOximeter pox;
17 uint32_t tsLastReport=0;
18 void onBeatDetected(){
19     ;
20 }
21 void callback(char *topic,byte *payload,unsigned int length)
22 {
23     Serial.print("Message arrived[");
24     Serial.print(topic);
25     Serial.print("]");
26     for(int i=0;i<length;i++)
27     {
28         Serial.print((char)payload[i]);
29     }
30     Serial.println();
31 }
32 void setup(){
33     Serial.begin(115200);
34     led_init();
35 }
```

Fig: 4.1 Coding 1

```

File Edit Sketch Tools Help
ESP32-WROOM-DA Mod...

ProjectOximeterHeartRateSensor_Final2.ino
30 Serial.println();
31 }
32 void setup(){
33   Serial.begin(115200);
34   lcd.init();
35   lcd.backlight();
36   ubidots.connectToWifi(WIFI_SSID,WIFI_PASS);
37   ubidots.setCallback(callback);
38   ubidots.setup();
39   ubidots.reconnect();
40   Serial.print("Intializing pulse oximeter..");
41   if(!pox.begin()){
42     Serial.println("FAILED");
43     for(;;)
44     {
45       ;
46     }
47   }else{
48     Serial.println("SUCESS");
49     digitalWrite(1,HIGH);
50   }
51   pox.setIRLedCurrent(MAX30100_LED_CURR_24mA);
52   pox.setOnBeatDetectedCallback(onBeatDetected);
53 }
54 void loop()
55 {
56   if(!ubidots.connected()){
57     ubidots.reconnect();
58   }
59   pox.update();
60   if (millis()-tsLastReport>REPORTING_PERIOD_MS){
61     ubidots.add(VARIABLE_LABEL1,pox.getHeartRate());
62     ubidots.publish(DEVICE_LABEL);
63   }
64 }

```

Fig: 4.2 Coding 2

```

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ESP32-WROOM-DA Mod...

ProjectOximeterHeartRateSensor_Final2.ino
51 pox.setIRLedCurrent(MAX30100_LED_CURR_24mA);
52 pox.setOnBeatDetectedCallback(onBeatDetected);
53 }
54 void loop()
55 {
56   if(!ubidots.connected()){
57     ubidots.reconnect();
58   }
59   pox.update();
60   if (millis()-tsLastReport>REPORTING_PERIOD_MS){
61     ubidots.add(VARIABLE_LABEL1,pox.getHeartRate());
62     ubidots.publish(DEVICE_LABEL);
63     ubidots.add(VARIABLE_LABEL2,pox.getSpO2());
64     ubidots.publish(DEVICE_LABEL);
65     int sensor = analogRead(A0);
66     ubidots.add(VARIABLE_LABEL3,sensor);
67     ubidots.publish(DEVICE_LABEL);
68     lcd.setCursor(0,0);
69     lcd.print("ECG:");
70     lcd.print(sensor);
71     lcd.setCursor(0,1);
72     lcd.print("BPM:");
73     lcd.print(pox.getHeartRate());
74     lcd.setCursor(10,0);
75     lcd.print("SpO2:");
76     lcd.setCursor(12,1);
77     lcd.print(pox.getSpO2());
78     lcd.setCursor(14,1);
79     lcd.print("%");
80     ubidots.loop();
81     tsLastReport=millis();
82   }
83 }

```

Fig: 4.3 Coding 3

CHAPTER 5

WORKING OF PROTOTYPE MODULE

In this chapter we have designed the working of prototype model along with their data analysis is shown how the micro-controller are synced interfaced with MAX30100 sensors, AD8232 ECG and I2C LCD in which the data is fetched and send to the web-server for visualization of the data here we have also used an alert SMS/Email system which notifies their concerned ones and also notifies the doctor to take immediate action.

5.1 CIRCUIT DIAGRAM OF PROTOTYPE MODEL

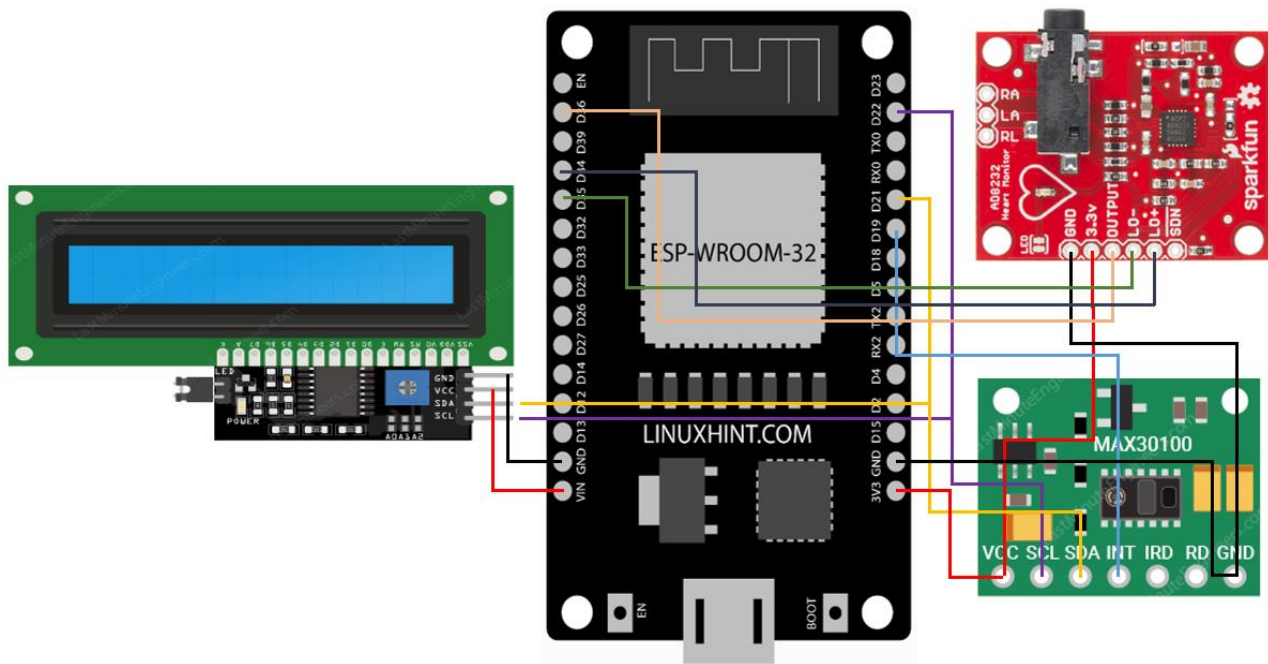


Fig: 5.1 E-Circuit Diagram

5.2 WORKING

The process behind the working of Health Monitoring system is end to end communication. Here are the steps of connection as shown in the Fig.5.1:-

1. ESP32 pins D19,D21 ,D22,3.3V,5V and two GND pins are used.

2. MAX30100 consist of 7 pins but for this project we require 5 pins -VIN with 3.3V, SCL with D21 of ESP32 ,SDA with D22,INT with D19 and GND with GND of ESP32 module.
3. I2C 16X2 LCD pins VIN is connected with 5V,GND with GND of ESP32 module, SDA and SCL pins are shared by MAX30100.
4. AD8232 consists of 6 pins but we require 5 pins -3.3V connected to 3.3V, OUTPUT- D36 (VP), LO- with D35, LO+ with D34 and GND - GND of ESP32 module.

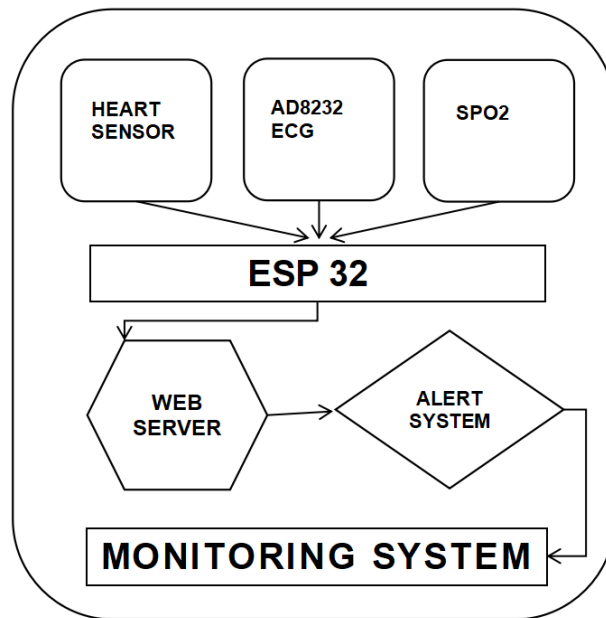


Fig: 5.2 END TO END COMMUNICATION

5.3 DATA VISUALIZATION

Prototype Model

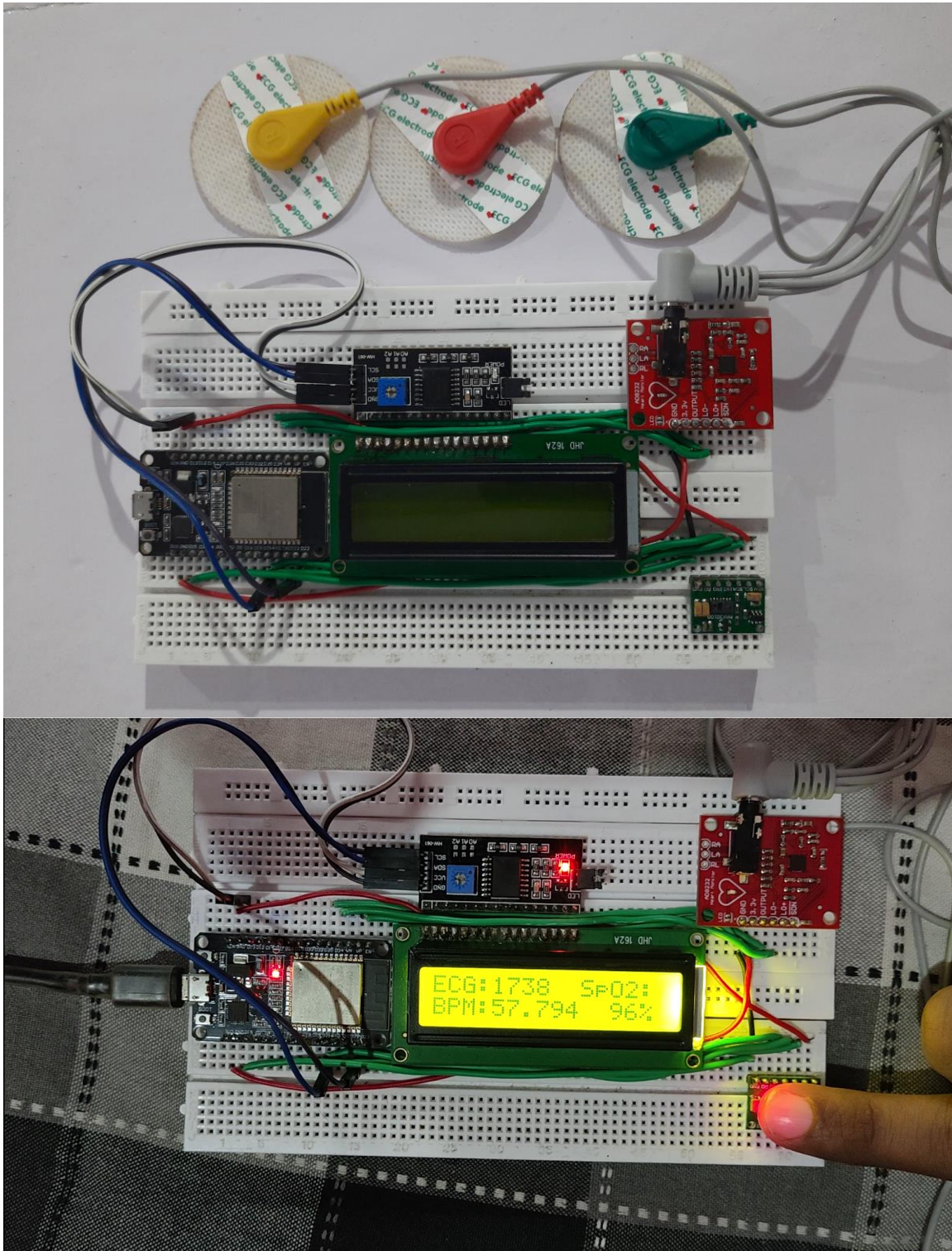


Fig: 5.3 Prototype Model

Web-Server Data Visualization

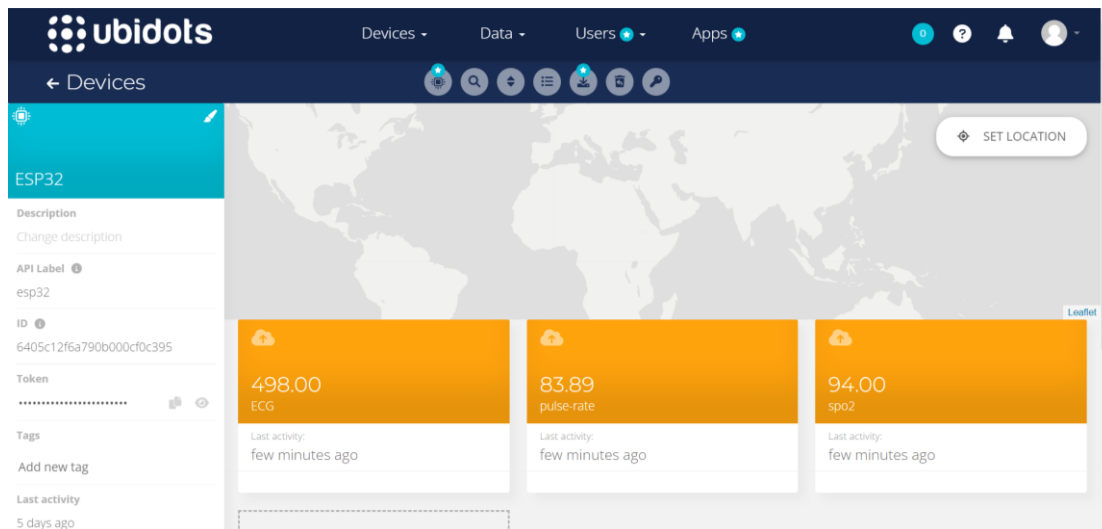


Fig: 5.3.2 Server Data



Fig: 5.3.4 BPM Chart



Fig: 5.3.6 SpO2 Level Chart

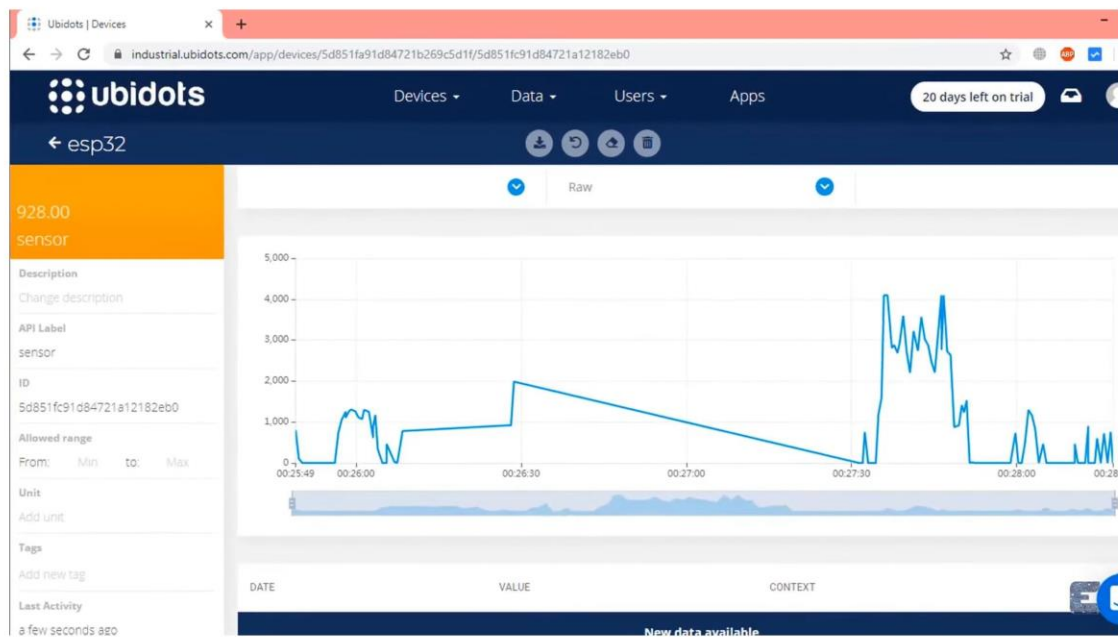


Fig: 5.3.7 ECG Level Chart

ALERT

Notifications Ubidots <service@ubidots.com> Mon, May 6, 4:10 PM
to me ▾
Hi there,
spo2 is back to normal. Current value is: 95.00.
Thank you.

Notifications Ubidots <service@ubidots.com> Mon, May 6, 4:10 PM
to me ▾
Hi there,
spo2 was 0.00 at 2024-05-06 10:40:47 +0000.
Thank you.

Notifications Ubidots <service@ubidots.com> Mon, May 6, 4:11 PM
to me ▾

Notifications Ubidots <service@ubidots.com> Mon, May 6, 4:11 PM
to me ▾
Hi there,
spo2 was 93.00 at 2024-05-06 10:41:02 +0000.
Thank you.

CHAPTER 6

MERITS AND DEMERITS

Remote Patient Monitoring systems are especially useful because they let the patients live their life while at the same time afford constant medical attention. The need for visiting the clinic/doctor is pushed to only deserving cases. Offline or online RPM devices are effective patient companions at all times. RPM systems can be used by even seemingly healthy people who may foresee health problems in the future. RPM systems will continue to evolve with increasing awareness, utilization and improving technologies.

The progress in bio medical engineering, science and technology paved way for new inventions and technologies. As we are moving towards miniaturization, handy electronic components are in need. New products and new technology are being invented. ARDUINO was found to be more compact, user friendly and less complex, which could readily be used in order to perform several tedious and repetitive tasks. Simulation is performed using Arduino software by placing appropriate sensors like temperature and heart beat rate for sensing the health condition and the results are analyzed under normal conditions and abnormality conditions.

This project can be further enhanced by sensing and displaying other vital statistics of a patient like ECG, blood pressure, glucose level etc. the other thing which is to add is presently we are monitoring the data in Arduino IDE in future we can monitor data in web page using internet of thing technology.

In future, a portable health monitoring system can be designed using Arduino.

Remote Patient Monitoring comes with a lot of promise and has evolved into a finer service than Telehealth. It is going to continue evolving with time in terms of growth and innovation. The following trends are foreseen in the future for Remote Patient Monitoring:

- Increased adoption of Remote Patient Monitoring in Practice Management and Research.
- Consolidation and convergence of technologies participating in RPM.
- Increasing patient participation in RPM adoption and utilization.

- Improvements in devices and technologies with increasing utilization and information share.
- Application of analytics leading to a better study of patient condition information and the related assessments.

RPM system could find applications in numerous healthcare cases some of which are quoted below:

Heart Patients:

Recovering heart patients could require continuous monitoring to assess the conditions and quick care/medication. Devices which include heart resynchronization therapy and pacemakers are especially useful in preventing problems for heart patients. The diagnostic software combined with the device capabilities could almost double-up as a doctor in the vicinity.

Senility or Dementia Problems:

Old age people often suffer from senility and forgetfulness; this could bring some problems such as losing the way or such people falling down at some places. The surveillance and assistance provided by RPM devices could help such people; if these are GPS enabled, they could even help track their locations.

Diabetes and Hypertension Control:

Diabetic or high BP patients require continuous monitoring and control of blood pressure and blood sugar; this can be achieved by an RPM system that not only monitors but could also provide alerts for controlling them. The diagnostic software could provide not only the condition but even advice the diet control or medication necessary to bring the blood sugar to acceptable levels.

Clinical Trials:

In specific clinical trials, which may require long-term observation of the subjects during and after the trials, this may be achieved with the RPM systems.

CHAPTER 7

CONCLUSION

7.1 SUMMARY

The major goal of the project was to design a system that could gather readings of the patient's numerous symptoms, analyse them in the cloud, and then alert the doctor or other concerned parties about the patient's health. This was achieved by developing a system that is implanted and relies on sensors to communicate the reading of significant indicators to cloud administrations provided by Ubidots Web-Server. The readings fetched are seen in the LCD display in the presence and the data can be shown by logic into the Ubidots Web-Server. The results of the data analysis performed on the model are shown in the above-mentioned figure. The sensor data are utilised to keep track of any changes in a patient's heart rate and oxygen saturation level, and the new status is relayed to the Ubidots Server using MQTT. Thus, the suggested system is capable of storing readings of several patient indicators including heart rate, ECG track and SpO2. It monitors the change in values and monitors abnormalities. These anomalies raise awareness among the medical staff, the treating physician, and the patient's family, which lessens the need for manual monitoring. To communicate the data to the cloud platform, the system employs MQTT communication. This messaging protocol aids the concerned parties and the doctor by transmitting the readings of significant patients' vital signs.

7.2 IMPACT ASSESSMENT

Application of IOT based Health Monitoring System working at 2 to 3.6V supply at max 5V using USB port.

Communication based application: This ESP32 Module micro-controller has several application in wireless communication. These are used in the fields of health care and communication and many other new prototype models. It uses a minimum of 2 to 3.6V supply to operate and also provide WiFi communication.

Medical applications: This micro-controller not only used in robotics but also used in medical purposes also. We have track the patient's pulses and oxygen level, along with AD8232 ECG sensors interfaced with the ESP32 micro-controller.

7.3 COST ANALYSIS

Since the project work is made a type of prototype, there is procurement cost required. According to Amazon, an e-commerce site, the cost of procurement of Jumper Wires, MAX30100 Pulse Oximeter, AD8232, ESP32 WiFi Module, PCB Board, Breadboard and I2C 16X2 LCD is 2262.81 Rs.



Fig: 7.3 Components

APPENDIX – COMPONENT LIST & DATA SHEETS

1 Overview

ESP32-WROOM-32 is a powerful, generic Wi-Fi + Bluetooth® + Bluetooth LE MCU module that targets a wide variety of applications, ranging from low-power sensor networks to the most demanding tasks, such as voice encoding, music streaming and MP3 decoding.

At the core of this module is the ESP32-D0WDQ6 chip*. The chip embedded is designed to be scalable and adaptive. There are two CPU cores that can be individually controlled, and the CPU clock frequency is adjustable from 80 MHz to 240 MHz. The chip also has a low-power coprocessor that can be used instead of the CPU to save power while performing tasks that do not require much computing power, such as monitoring of peripherals. ESP32 integrates a rich set of peripherals, ranging from capacitive touch sensors, SD card interface, Ethernet, high-speed SPI, UART, I2S, and I2C.

Note:

* For details on the part numbers of the ESP32 family of chips, please refer to the document [ESP32 Datasheet](#).

The integration of Bluetooth, Bluetooth LE and Wi-Fi ensures that a wide range of applications can be targeted, and that the module is all-around: using Wi-Fi allows a large physical range and direct connection to the Internet through a Wi-Fi router, while using Bluetooth allows the user to conveniently connect to the phone or broadcast low energy beacons for its detection. The sleep current of the ESP32 chip is less than 5 μ A, making it suitable for battery powered and wearable electronics applications. The module supports a data rate of up to 150 Mbps, and 20 dBm output power at the antenna to ensure the widest physical range. As such the module does offer industry-leading specifications and the best performance for electronic integration, range, power consumption, and connectivity.

The operating system chosen for ESP32 is freeRTOS with LwIP; TLS 1.2 with hardware acceleration is built in as well. Secure (encrypted) over the air (OTA) upgrade is also supported, so that users can upgrade their products even after their release, at minimum cost and effort.

Table 1 provides the specifications of ESP32-WROOM-32.

Table 1: ESP32-WROOM-32 Specifications

Categories	Items	Specifications
Certification	RF certification	See certificates for ESP32-WROOM-32
	Wi-Fi certification	Wi-Fi Alliance
	Bluetooth certification	BQB
	Green certification	RoHS/REACH
Test	Reliability	HTOL/HTSL/uHAST/TCT/ESD
Wi-Fi	Protocols	802.11 b/g/n (802.11n up to 150 Mbps)
		A-MPDU and A-MSDU aggregation and 0.4 μ s guard interval support
	Center frequency range of operating channel	2412 ~ 2484 MHz
Bluetooth	Protocols	Bluetooth v4.2 BR/EDR and Bluetooth LE specification
	Radio	NZIF receiver with -97 dBm sensitivity
		Class-1, class-2 and class-3 transmitter
		AFH

Categories	Items	Specifications
	Audio	CVSD and SBC
Hardware	Module interfaces	SD card, UART, SPI, SDIO, I2C, LED PWM, Motor PWM, I2S, IR, pulse counter, GPIO, capacitive touch sensor, ADC, DAC, Two-Wire Automotive Interface (TWAI®), compatible with ISO11898-1 (CAN Specification 2.0)
	Integrated crystal	40 MHz crystal
	Integrated SPI flash	4 MB
	Operating voltage/Power supply	3.0 V ~ 3.6 V
	Operating current	Average: 80 mA
	Minimum current delivered by power supply	500 mA
	Recommended operating ambient temperature range	-40 °C ~ +85 °C
	Package size	18 mm × 25.5 mm × 3.10 mm
	Moisture sensitivity level (MSL)	Level 3

2 Pin Definitions

2.1 Pin Layout

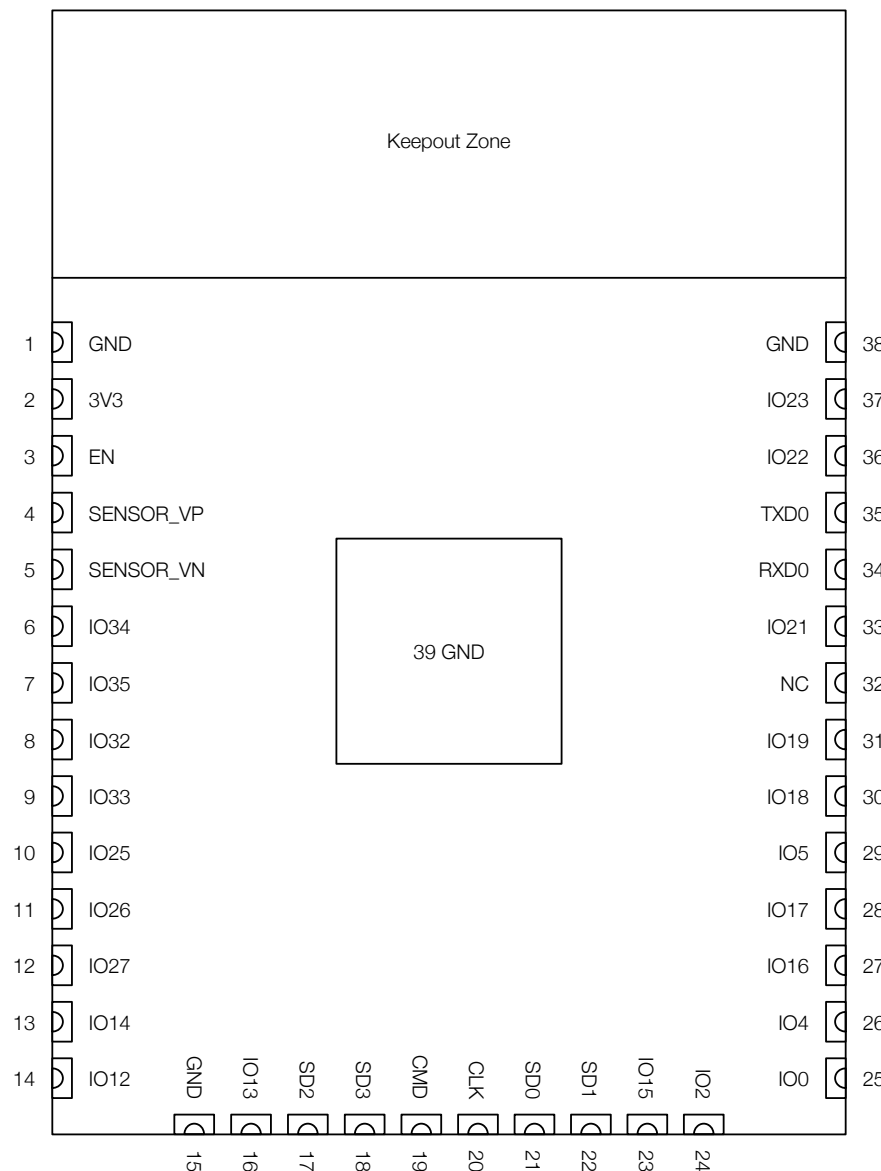


Figure 1: ESP32-WROOM-32 Pin Layout (Top View)

2.2 Pin Description

ESP32-WROOM-32 has 38 pins. See pin definitions in Table 2.

Table 2: Pin Definitions

Name	No.	Type	Function
GND	1	P	Ground
3V3	2	P	Power supply
EN	3	I	Module-enable signal. Active high.

Name	No.	Type	Function
SENSOR_VP	4	I	GPIO36, ADC1_CH0, RTC_GPIO0
SENSOR_VN	5	I	GPIO39, ADC1_CH3, RTC_GPIO3
IO34	6	I	GPIO34, ADC1_CH6, RTC_GPIO4
IO35	7	I	GPIO35, ADC1_CH7, RTC_GPIO5
IO32	8	I/O	GPIO32, XTAL_32K_P (32.768 kHz crystal oscillator input), ADC1_CH4, TOUCH9, RTC_GPIO9
IO33	9	I/O	GPIO33, XTAL_32K_N (32.768 kHz crystal oscillator output), ADC1_CH5, TOUCH8, RTC_GPIO8
IO25	10	I/O	GPIO25, DAC_1, ADC2_CH8, RTC_GPIO6, EMAC_RXD0
IO26	11	I/O	GPIO26, DAC_2, ADC2_CH9, RTC_GPIO7, EMAC_RXD1
IO27	12	I/O	GPIO27, ADC2_CH7, TOUCH7, RTC_GPIO17, EMAC_RX_DV
IO14	13	I/O	GPIO14, ADC2_CH6, TOUCH6, RTC_GPIO16, MTMS, HSPICLK, HS2_CLK, SD_CLK, EMAC_TXD2
IO12	14	I/O	GPIO12, ADC2_CH5, TOUCH5, RTC_GPIO15, MTDI, HSPIQ, HS2_DATA2, SD_DATA2, EMAC_TXD3
GND	15	P	Ground
IO13	16	I/O	GPIO13, ADC2_CH4, TOUCH4, RTC_GPIO14, MTCK, HSPID, HS2_DATA3, SD_DATA3, EMAC_RX_ER
SHD/SD2*	17	I/O	GPIO9, SD_DATA2, SPIHD, HS1_DATA2, U1RXD
SWP/SD3*	18	I/O	GPIO10, SD_DATA3, SPIWP, HS1_DATA3, U1TXD
SCS/CMD*	19	I/O	GPIO11, SD_CMD, SPICS0, HS1_CMD, U1RTS
SCK/CLK*	20	I/O	GPIO6, SD_CLK, SPICLK, HS1_CLK, U1CTS
SDO/SD0*	21	I/O	GPIO7, SD_DATA0, SPIQ, HS1_DATA0, U2RTS
SDI/SD1*	22	I/O	GPIO8, SD_DATA1, SPID, HS1_DATA1, U2CTS
IO15	23	I/O	GPIO15, ADC2_CH3, TOUCH3, MTDO, HSPICS0, RTC_GPIO13, HS2_CMD, SD_CMD, EMAC_RXD3
IO2	24	I/O	GPIO2, ADC2_CH2, TOUCH2, RTC_GPIO12, HSPIWP, HS2_DATA0, SD_DATA0
IO0	25	I/O	GPIO0, ADC2_CH1, TOUCH1, RTC_GPIO11, CLK_OUT1, EMAC_TX_CLK
IO4	26	I/O	GPIO4, ADC2_CH0, TOUCH0, RTC_GPIO10, HSPICLK, HS2_DATA1, SD_DATA1, EMAC_TX_ER
IO16	27	I/O	GPIO16, HS1_DATA4, U2RXD, EMAC_CLK_OUT
IO17	28	I/O	GPIO17, HS1_DATA5, U2TXD, EMAC_CLK_OUT_180
IO5	29	I/O	GPIO5, VSPICS0, HS1_DATA6, EMAC_RX_CLK
IO18	30	I/O	GPIO18, VSPICLK, HS1_DATA7
IO19	31	I/O	GPIO19, VSPIQ, U0CTS, EMAC_TXD0
NC	32	-	-
IO21	33	I/O	GPIO21, VSPIHD, EMAC_TX_EN
RXD0	34	I/O	GPIO3, U0RXD, CLK_OUT2
TXD0	35	I/O	GPIO1, U0TXD, CLK_OUT3, EMAC_RXD2
IO22	36	I/O	GPIO22, VSPIWP, U0RTS, EMAC_TXD1
IO23	37	I/O	GPIO23, VSPID, HS1_STROBE
GND	38	P	Ground

General Description

The MAX30100 is an integrated pulse oximetry and heart-rate monitor sensor solution. It combines two LEDs, a photodetector, optimized optics, and low-noise analog signal processing to detect pulse oximetry and heart-rate signals.

The MAX30100 operates from 1.8V and 3.3V power supplies and can be powered down through software with negligible standby current, permitting the power supply to remain connected at all times.

Applications

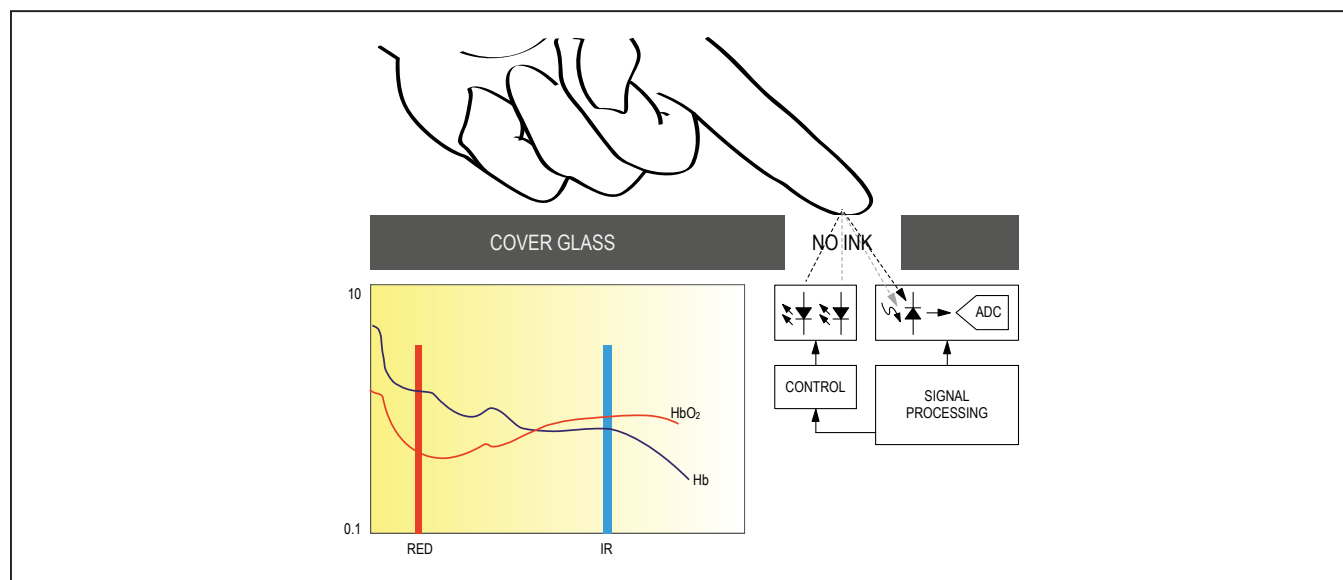
- Wearable Devices
- Fitness Assistant Devices
- Medical Monitoring Devices

Benefits and Features

- Complete Pulse Oximeter and Heart-Rate Sensor Solution Simplifies Design
 - Integrated LEDs, Photo Sensor, and High-Performance Analog Front -End
 - Tiny 5.6mm x 2.8mm x 1.2mm 14-Pin Optically Enhanced System-in-Package
- Ultra-Low-Power Operation Increases Battery Life for Wearable Devices
 - Programmable Sample Rate and LED Current for Power Savings
 - Ultra-Low Shutdown Current (0.7 μ A, typ)
- Advanced Functionality Improves Measurement Performance
 - High SNR Provides Robust Motion Artifact Resilience
 - Integrated Ambient Light Cancellation
 - High Sample Rate Capability
 - Fast Data Output Capability

Ordering Information appears at end of data sheet.

System Block Diagram



Absolute Maximum Ratings

V_{DD} to GND -0.3V to +2.2V
 GND to PGND -0.3V to +0.3V
 x_DRV , x_LED+ to PGND -0.3V to +6.0V
 All Other Pins to GND -0.3V to +6.0V
 Output Short-Circuit Current Duration Continuous
 Continuous Input Current into Any Terminal $\pm 20\text{mA}$

Continuous Power Dissipation ($T_A = +70^\circ\text{C}$)
 OESIP (derate 5.8mW/ $^\circ\text{C}$ above $+70^\circ\text{C}$) 464mW
 Operating Temperature Range -40°C to $+85^\circ\text{C}$
 Soldering Temperature (reflow) $+260^\circ\text{C}$
 Storage Temperature Range -40°C to $+105^\circ\text{C}$

Package Thermal Characteristics (Note 1)

OESIP

Junction-to-Ambient Thermal Resistance (θ_{JA}) 150°C/W
 Junction-to-Case Thermal Resistance (θ_{JC}) 170°C/W

Note 1: Package thermal resistances were obtained using the method described in JEDEC specification JESD51-7, using a four-layer board. For detailed information on package thermal considerations, refer to www.maximintegrated.com/thermal-tutorial.

Electrical Characteristics

($V_{DD} = 1.8\text{V}$, $V_{IR_LED+} = V_{R_LED+} = 3.3\text{V}$, $T_A = +25^\circ\text{C}$, min/max are from $T_A = -40^\circ\text{C}$ to $+85^\circ\text{C}$, unless otherwise noted.) (Note 2)

PARAMETER	SYMBOL	CONDITIONS	MIN	TYP	MAX	UNITS
POWER SUPPLY						
Power-Supply Voltage	V_{DD}	Guaranteed by RED and IR count tolerance	1.7	1.8	2.0	V
LED Supply Voltage (R_LED+ or IR_LED+ to PGND)	V_{LED+}	Guaranteed by PSRR of LED Driver	3.1	3.3	5.0	V
Supply Current	I_{DD}	SpO ₂ and heart rate modes, PW = 200 μs , 50sps		600	1200	μA
		Heart rate only mode, PW = 200 μs , 50sps		600	1200	
Supply Current in Shutdown	I_{SHDN}	$T_A = +25^\circ\text{C}$, MODE = 0x80		0.7	10	μA
SENSOR CHARACTERISTICS						
ADC Resolution				14		bits
Red ADC Count (Note 3)	RED_C	Propriety ATE setup RED_PA = 0x05, LED_PW = 0x00, SPO2_SR = 0x07, $T_A = +25^\circ\text{C}$	23,000	26,000	29,000	Counts
IR ADC Count (Note 3)	IR_C	Propriety ATE setup IR_PA = 0x09, LED_PW = 0x00, SPO2_SR = 0x07, $T_A = +25^\circ\text{C}$	23,000	26,000	29,000	Counts
Dark Current Count	DC_C	RED_PA = IR_PA = 0x00, LED_PW = 0x03, SPO2_SR = 0x01		0	3	Counts
DC Ambient Light Rejection (Note 4)	ALR	Number of ADC counts with finger on sensor under direct sunlight (100K lux) LED_PW = 0x03, SPO2_SR = 0x01	RED LED	0		Counts
			IR LED	0		

Electrical Characteristics (continued)(V_{DD} = 1.8V, V_{IR_LED+} = V_{R_LED+} = 3.3V, T_A = +25°C, min/max are from T_A = -40°C to +85°C, unless otherwise noted.) (Note 2)

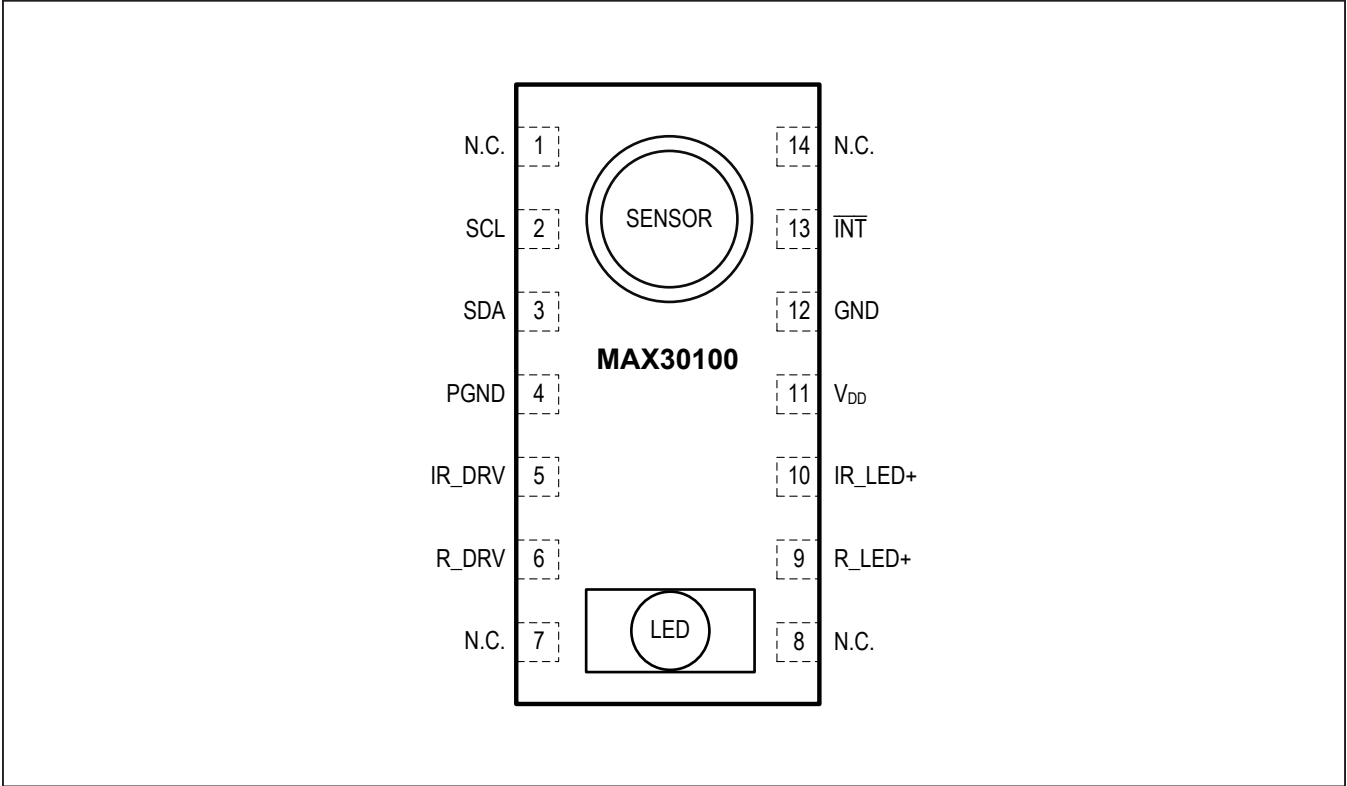
PARAMETER	SYMBOL	CONDITIONS	MIN	TYP	MAX	UNITS
IR ADC Count—PSRR (V _{DD})	PSRR _{VDD}	Propriety ATE setup 1.7V < V _{DD} < 2.0V, LED_PW = 0x03, SPO2_SR = 0x01, IR_PA = 0x09, IR_PA = 0x05, T _A = +25°C		0.25	2	%
		Frequency = DC to 100kHz, 100mV _{P-P}		10		LSB
RED/IR ADC Count—PSRR (X_LED+)	PSRR _{LED}	Propriety ATE setup 3.1V < X_LED+ < 5V, LED_PW = 0x03, SPO2_SR = 0x01, IR_PA = 0x09, IR_PA = 0x05, T _A = +25°C		0.05	2	%
		Frequency = DC to 100kHz, 100mV _{P-P}		10		LSB
ADC Integration Time	INT	LED_PW = 0x00		200		μs
		LED_PW = 0x03		1600		μs
IR LED CHARACTERISTICS (Note 4)						
LED Peak Wavelength	λ _P	I _{LED} = 20mA, T _A = +25°C	870	880	900	nm
Full Width at Half Max	Δλ	I _{LED} = 20mA, T _A = +25°C		30		nm
Forward Voltage	V _F	I _{LED} = 20mA, T _A = +25°C		1.4		V
Radiant Power	P _O	I _{LED} = 20mA, T _A = +25°C		6.5		mW
RED LED CHARACTERISTICS (Note 4)						
LED Peak Wavelength	λ _P	I _{LED} = 20mA, T _A = +25°C	650	660	670	nm
Full Width at Half Max	Δλ	I _{LED} = 20mA, T _A = +25°C		20		nm
Forward Voltage	V _F	I _{LED} = 20mA, T _A = +25°C		2.1		V
Radiant Power	P _O	I _{LED} = 20mA, T _A = +25°C		9.8		mW
TEMPERATURE SENSOR						
Temperature ADC Acquisition Time	T _T	T _A = +25°C		29		ms
Temperature Sensor Accuracy	T _A	T _A = +25°C		±1		°C
Temperature Sensor Minimum Range	T _{MIN}			-40		°C
Temperature Sensor Maximum Range	T _{MAX}			85		°C

Electrical Characteristics (continued)(V_{DD} = 1.8V, V_{IR_LED+} = V_{R_LED+} = 3.3V, T_A = +25°C, min/max are from T_A = -40°C to +85°C, unless otherwise noted.) (Note 2)

PARAMETER	SYMBOL	CONDITIONS	MIN	TYP	MAX	UNITS
DIGITAL CHARACTERISTICS (SDA, SDA, INT)						
Output Low Voltage SDA, INT	V _{OL}	I _{SINK} = 6mA			0.4	V
I ² C Input Voltage Low	V _{IL_I2C}	SDA, SCL			0.4	V
I ² C Input Voltage High	V _{IH_I2C}	SDA, SCL	1.4			V
Input Hysteresis	V _{HYS}	SDA, SCL		200		mV
Input Capacitance	C _{IN}	SDA, SCL		10		pF
Input Leakage Current	I _{IN}	V _{IN} = 0V, T _A = +25°C (SDA, SCL, INT)		0.01	1	μA
		V _{IN} = 5.5V, T _A = +25°C (SDA, SCL, INT)		0.01	1	μA
I ² C TIMING CHARACTERISTICS (SDA, SDA, INT)						
I ² C Write Address				AE		Hex
I ² C Read Address				AF		Hex
Serial Clock Frequency	f _{SCL}		0		400	kHz
Bus Free Time Between STOP and START Conditions	t _{BUF}		1.3			μs
Hold Time (Repeated) START Condition	t _{HD,START}		0.6			μs
SCL Pulse-Width Low	t _{LOW}		1.3			μs
SCL Pulse-Width High	t _{HIGH}		0.6			μs
Setup Time for a Repeated START Condition	t _{SU,START}		0.6			μs
Data Hold Time	t _{HD,DAT}		0		900	ns
Data Setup Time	t _{SU,DAT}		100			ns
Setup Time for STOP Condition	t _{SU,STOP}		0.6			μs
Pulse Width of Suppressed Spike	t _{SP}		0		50	ns
Bus Capacitance	C _B				400	pF
SDA and SCL Receiving Rise Time	t _R		20 + 0.1C _B		300	ns
SDA and SCL Receiving Fall Time	t _{RF}		20 + 0.1C _B		300	ns
SDA Transmitting Fall Time	t _{TF}		20 + 0.1C _B		300	ns

Note 2: All devices are 100% production tested at T_A = +25°C. Specifications over temperature limits are guaranteed by Maxim Integrated's bench or proprietary automated test equipment (ATE) characterization.**Note 3:** Specifications are guaranteed by Maxim Integrated's bench characterization and by 100% production test using proprietary ATE setup and conditions.**Note 4:** For design guidance only. Not production tested.

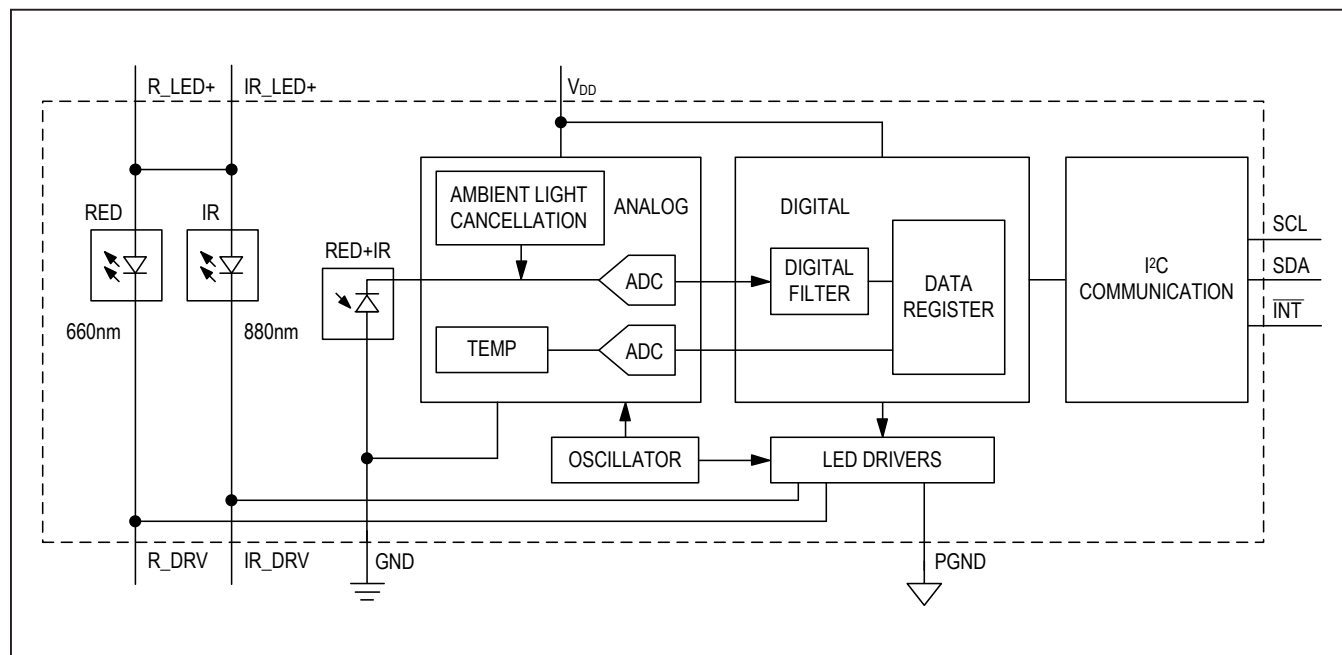
Pin Configuration



Pin Description

PIN	NAME	FUNCTION
1, 7, 8, 14	N.C.	No Connection. Connect to PCB Pad for Mechanical Stability.
2	SCL	I ² C Clock Input
3	SDA	I ² C Clock Data, Bidirectional (Open-Drain)
4	PGND	Power Ground of the LED Driver Blocks
5	IR_DRV	IR LED Cathode and LED Driver Connection Point. Leave floating in circuit.
6	R_DRV	Red LED Cathode and LED Driver Connection Point. Leave floating in circuit.
9	R_LED+	Power Supply (Anode Connection) for Red LED. Bypass to PGND for best performance. Connected to IR_LED+ internally.
10	IR_LED+	Power Supply (Anode Connection) for IR LED. Bypass to PGND for best performance. Connected to R_LED+ internally.
11	V _{DD}	Analog Power Supply Input. Bypass to GND for best performance.
12	GND	Analog Ground
13	INT	Active-Low Interrupt (Open-Drain)

Functional Diagram



Detailed Description

The MAX30100 is a complete pulse oximetry and heart-rate sensor system solution designed for the demanding requirements of wearable devices. The MAX30100 provides very small total solution size without sacrificing optical or electrical performance. Minimal external hardware components are needed for integration into a wearable device.

The MAX30100 is fully configurable through software registers, and the digital output data is stored in a 16-deep FIFO within the device. The FIFO allows the MAX30100 to be connected to a microcontroller or microprocessor on a shared bus, where the data is not being read continuously from the device's registers.

SpO₂ Subsystem

The SpO₂ subsystem in the MAX30100 is composed of ambient light cancellation (ALC), 16-bit sigma delta ADC, and proprietary discrete time filter.

The SpO₂ ADC is a continuous time oversampling sigma delta converter with up to 16-bit resolution. The ADC output data rate can be programmed from 50Hz to 1kHz. The

MAX30100 includes a proprietary discrete time filter to reject 50Hz/60Hz interference and low-frequency residual ambient noise.

Temperature Sensor

The MAX30100 has an on-chip temperature sensor for (optionally) calibrating the temperature dependence of the SpO₂ subsystem.

The SpO₂ algorithm is relatively insensitive to the wavelength of the IR LED, but the red LED's wavelength is critical to correct interpretation of the data. The temperature sensor data can be used to compensate the SpO₂ error with ambient temperature changes.

LED Driver

The MAX30100 integrates red and IR LED drivers to drive LED pulses for SpO₂ and HR measurements. The LED current can be programmed from 0mA to 50mA (typical only) with proper supply voltage. The LED pulse width can be programmed from 200μs to 1.6ms to optimize measurement accuracy and power consumption based on use cases.

SPECIFICATIONS

$V_S = 3\text{ V}$, $V_{REF} = 1.5\text{ V}$, $V_{CM} = 1.5\text{ V}$, $T_A = 25^\circ\text{C}$, operating temperature (T_{OPR}) = -40°C to $+105^\circ\text{C}$ for the W grade, FR = low, SDN = high, and AC/DC = low, unless otherwise noted.

Table 1.

Parameter	Symbol	Test Conditions/ Comments	Min	A Grade Typ	Max	Min	W Grade Typ	Max	Unit
INSTRUMENTATION AMPLIFIER									
Common-Mode Rejection Ratio, DC to 60 Hz	CMRR	$V_{CM} = 0.35\text{ V to }2.85\text{ V}$, $V_{DIFF} = 0\text{ V}$	80	86					dB
		T_{OPR}				80			dB
		$V_{CM} = 0.35\text{ V to }2.85\text{ V}$, $V_{DIFF} = \pm 0.3\text{ V}$		80			80		dB
Power Supply Rejection Ratio	PSRR	$V_S = 2.0\text{ V to }3.5\text{ V}$	76	90					dB
		T_{OPR}				76	90		dB
Offset Voltage (RTI) Instrumentation Amplifier Inputs	V_{OS}			3	8				mV
		T_{OPR}				3	8		mV
				5	50		5	50	μV
DC Blocking Input ¹									
Average Offset Drift Instrumentation Amplifier Inputs	T_{OPR}			10			10		$\mu\text{V}/^\circ\text{C}$
									$\mu\text{V}/^\circ\text{C}$
				0.05			0.05		$\mu\text{V}/^\circ\text{C}$
DC Blocking Input ¹									
Input Bias Current	I_B			50	200		50	200	pA
		$T_A = 0^\circ\text{C to }70^\circ\text{C}$		1					nA
		T_{OPR}						3.75	nA
Input Offset Current	I_{OS}			25	100		25	100	pA
		$T_A = 0^\circ\text{C to }70^\circ\text{C}$		1					nA
		T_{OPR}						1	nA
Input Impedance									
				10 7.5			10 7.5		$\text{G}\Omega \text{pF}$
Common Mode				5 15			5 15		$\text{G}\Omega \text{pF}$
Input Voltage Noise (RTI)									
		$f = 1\text{ kHz}$		100			100		$\text{nV}/\sqrt{\text{Hz}}$
		$f = 0.1\text{ Hz to }10\text{ Hz}$		12			12		$\mu\text{V p-p}$
Peak-to-Peak Voltage Noise									
		$f = 0.5\text{ Hz to }40\text{ Hz}$		14			14		$\mu\text{V p-p}$
		$T_A = 0^\circ\text{C to }70^\circ\text{C}$			$+V_S$				V
Input Voltage Range	V_{DIFF}	T_{OPR}	0.2			0.2		$+V_S$	V
		T_{OPR}	-300		+300	-300		+300	mV
Output									
Output Swing		$R_L = 50\text{ k}\Omega$	0.1		$+V_S - 0.1$				V
		T_{OPR}				0.1		$+V_S - 0.1$	V
Short-Circuit Current	I_{OUT}			6.3			6.3		mA
Gain	A_V			100			100		V/V
Gain Error		$V_{DIFF} = 0\text{ V}$		0.4			0.4		%
		$V_{DIFF} = -300\text{ mV to }+300\text{ mV}$		1	3.5			3.5	%
		T_{OPR}						7.9	%
Average Gain Drift		$T_A = 0^\circ\text{C to }70^\circ\text{C}$		12					$\text{ppm}/^\circ\text{C}$
		T_{OPR}					12		$\text{ppm}/^\circ\text{C}$
Bandwidth	BW			2			2		kHz
RFI Filter Cutoff (Each Input)				1			1		MHz

Parameter	Symbol	Test Conditions/ Comments	Min	A Grade Typ	Max	Min	W Grade Typ	Max	Unit
OPERATIONAL AMPLIFIER (A1)									
Offset Voltage	V _{OS}	T _{OPR}		1	5		1	5	mV
Average TC		T _A = 0°C to 70°C		5					μV/°C
Input Bias Current	I _B	T _{OPR}		100			5		μV/°C
		T _A = 0°C to 70°C		1			100		pA
Input Offset Current	I _{OS}	T _{OPR}		100			2.5		nA
		T _A = 0°C to 70°C		1			100		nA
		T _{OPR}					1		nA
Input Voltage Range			0.1		+V _S − 0.1	0.1		+V _S − 0.1	V
Common-Mode Rejection Ratio	CMRR	V _{CM} = 0.5 V to 2.5 V		100			100		dB
Power Supply Rejection Ratio	PSRR			100			100		dB
Large Signal Voltage Gain	A _{VO}			110			110		dB
Output Voltage Range		R _L = 50 kΩ	0.1		+V _S − 0.1				V
		T _{OPR}				0.1		+V _S − 0.1	V
Short-Circuit Current Limit	I _{OUT}			12			12		mA
Gain Bandwidth Product	GBP			100			100		kHz
Slew Rate	SR			0.02			0.02		V/μs
Voltage Noise Density (RTI)	e _n	f = 1 kHz		60			60		nV/√Hz
Peak-to-Peak Voltage Noise (RTI)	e _{n p-p}	f = 0.1 Hz to 10 Hz		6			6		μV p-p
		f = 0.5 Hz to 40 Hz		8			8		μV p-p
RIGHT LEG DRIVE AMPLIFIER (A2)									
Output Swing		R _L = 50 kΩ	0.1		+V _S − 0.1	0.1		+V _S − 0.1	V
Short-Circuit Current	I _{OUT}			11			11		mA
Integrator Input Resistor			120	150	180	120	150	180	kΩ
Gain Bandwidth Product	GDP			100			100		kHz
REFERENCE BUFFER (A3)									
Offset Error	V _{OS}	R _L > 50 kΩ		1			1		mV
Input Bias Current	I _B			100			100		pA
Short-Circuit Current Limit	I _{OUT}			12			12		mA
Voltage Range		R _L = 50 kΩ	0.1		+V _S − 0.7				V
		T _{OPR}				0.1		+V _S − 0.7	V
DC LEADS OFF COMPARATORS									
Threshold Voltage				+V _S − 0.5			+V _S − 0.5		V
Hysteresis				60			60		mV
Propagation Delay				0.5			0.5		μs
AC LEADS OFF DETECTOR									
Square Wave Frequency	f _{AC}		50	100	175	50	100	175	kHz
Square Wave Amplitude	I _{AC}			200			200		nA p-p
Impedance Threshold		Between +IN and −IN	10	20		10	20		MΩ
Detection Delay				110			110		μs
FAST RESTORE CIRCUIT									
Switches		S1 and S2							
On Resistance	R _{ON}		8	10	12	8	10	12	kΩ
Off Leakage				100			100		pA
Window Comparator									
Threshold Voltage		From either rail		50			50		mV
Propagation Delay				2			2		μs
Switch Timing Characteristics									
Feedback Recovery Switch On Time	t _{SW1}			110			110		ms
Filter Recovery Switch On Time	t _{SW2}			55			55		ms
Fast Restore Reset	t _{RST}			2			2		μs

Parameter	Symbol	Test Conditions/ Comments	Min	A Grade Typ	Max	Min	W Grade Typ	Max	Unit	
LOGIC INTERFACE										
Input Characteristics										
Input Voltage (AC/ $\overline{DC}$ and FR)										
Low	V _{IL}	LOD+ and LOD– terminals		1.24			1.24		V	
High	V _{IH}			1.35			1.35		V	
Input Voltage ($\overline{SDN}$)										
Low	V _{IL}			2.1			2.1		V	
High	V _{IH}			0.5			0.5		V	
Output Characteristics										
Output Voltage										
Low	V _{OL}				0.05			0.05		V
High	V _{OH}			2.95			2.95		V	
SYSTEM SPECIFICATIONS										
Quiescent Supply Current		T _A = 0°C to 70°C		170	230		170	230	μA	
		T _{OPR}		210					μA	
Shutdown Current		T _A = 0°C to 70°C		40	500			270	μA	
		T _{OPR}		100		40	500		nA	
Supply Range			2.0		3.5	2.0		612	nA	
Specified Temperature Range			0		+70	–40			V	
Operational Temperature Range			–40		+85	–40		+105	°C	

¹ Offset referred to the input of the instrumentation amplifier inputs. See the Input Referred Offsets section for additional information.

ABSOLUTE MAXIMUM RATINGS

Table 2.

Parameter	Rating
Supply Voltage	3.6 V
Output Short-Circuit Current Duration	Indefinite
Maximum Voltage, Any Terminal ¹	+V _S + 0.3 V
Minimum Voltage, Any Terminal ¹	−0.3 V
Storage Temperature Range	−65°C to +125°C
Operating Temperature Range	
AD8232ACPZ	−40°C to +85°C
AD8232WACSZ	−40°C to +105°C
Maximum Junction Temperature	140°C
θ _{JA} Thermal Impedance ²	48°C/W
θ _{JC} Thermal Impedance	4.4°C/W
ESD Rating	
Human Body Model (HBM)	8 kV
Charged Device Model (FICDM)	1.25 kV
Machine Model (MM)	200 V

¹ This level or the maximum specified supply voltage, whichever is the lesser, indicates the superior voltage limit for any terminal. If input voltages beyond the specified minimum or maximum voltages are expected, place resistors in series with the inputs to limit the current to less than 5 mA.

² θ_{JA} is specified for a device in free air on a 4-layer JEDEC board.

Stresses at or above those listed under Absolute Maximum Ratings may cause permanent damage to the product. This is a stress rating only; functional operation of the product at these or any other conditions above those indicated in the operational section of this specification is not implied. Operation beyond the maximum operating conditions for extended periods may affect product reliability.

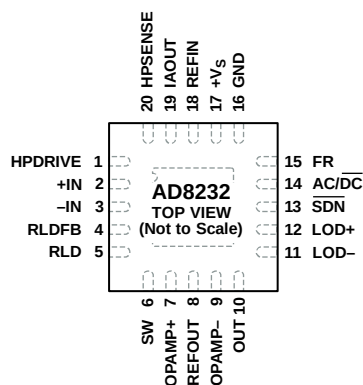
ESD CAUTION



ESD (electrostatic discharge) sensitive device.

Charged devices and circuit boards can discharge without detection. Although this product features patented or proprietary protection circuitry, damage may occur on devices subjected to high energy ESD. Therefore, proper ESD precautions should be taken to avoid performance degradation or loss of functionality.

PIN CONFIGURATION AND FUNCTION DESCRIPTIONS



NOTES
1. CONNECT THE EXPOSED PAD TO GND OR LEAVE UNCONNECTED.

10866-002

Figure 2. Pin Configuration

Table 3. Pin Function Descriptions

Pin No.	Mnemonic	Description
1	HPDRIVE	High-Pass Driver Output. Connect HPDRIVE to the capacitor in the first high-pass filter. The AD8232 drives this pin to keep HPSENSE at the same level as the reference voltage.
2	+IN	Instrumentation Amplifier Positive Input. +IN is typically connected to the left arm (LA) electrode.
3	-IN	Instrumentation Amplifier Negative Input. -IN is typically connected to the right arm (RA) electrode.
4	RLDFB	Right Leg Drive Feedback Input. RLDFB is the feedback terminal for the right leg drive circuit.
5	RLD	Right Leg Drive Output. Connect the driven electrode (typically, right leg) to the RLD pin.
6	SW	Fast Restore Switch Terminal. Connect this terminal to the output of the second high-pass filter.
7	OPAMP+	Operational Amplifier Noninverting Input.
8	REFOUT	Reference Buffer Output. The instrumentation amplifier output is referenced to this potential. Use REFOUT as a virtual ground for any point in the circuit that needs a signal reference.
9	OPAMP-	Operational Amplifier Inverting Input.
10	OUT	Operational Amplifier Output. The fully conditioned heart rate signal is present at this output. OUT can be connected to the input of an ADC.
11	LOD-	Leads Off Comparator Output. In dc leads off detection mode, LOD- is high when the electrode to -IN is disconnected, and it is low when connected. In ac leads off detection mode, LOD- is always low.
12	LOD+	Leads Off Comparator Output. In dc leads off detection mode, LOD+ is high when the +IN electrode is disconnected, and it is low when connected. In ac leads off detection mode, LOD+ is high when either the -IN or +IN electrode is disconnected, and it is low when both electrodes are connected.
13	$\overline{\text{SDN}}$	Shutdown Control Input. Drive $\overline{\text{SDN}}$ low to enter the low power shutdown mode.
14	$\overline{\text{AC/DC}}$	Leads Off Mode Control Input. Drive the $\overline{\text{AC/DC}}$ pin low for dc leads off mode. Drive the $\overline{\text{AC/DC}}$ pin high for ac leads off mode.
15	FR	Fast Restore Control Input. Drive FR high to enable fast recovery mode; otherwise, drive it low.
16	GND	Power Supply Ground.
17	+Vs	Power Supply Terminal.
18	REFIN	Reference Buffer Input. Use REFIN, a high impedance input terminal, to set the level of the reference buffer.
19	IAOUT	Instrumentation Amplifier Output Terminal.
20	HPSENSE	High-Pass Sense Input for Instrumentation Amplifier. Connect HPSENSE to the junction of R and C that sets the corner frequency of the dc blocking circuit.
	EP	Exposed Pad. Connect the exposed pad to GND or leave it unconnected.

I2C Serial Interface 1602 LCD Module

This is I2C interface 16x2 LCD display module, a high-quality 2 line 16 character LCD module with on-board contrast control adjustment, backlight and I2C communication interface. For Arduino beginners, no more cumbersome and complex LCD driver circuit connection. The real significance advantages of this I2C Serial LCD module will simplify the circuit connection, save some I/O pins on Arduino board, simplified firmware development with widely available Arduino library.



SKU: [DSP-1182](#)

Brief Data:

- Compatible with Arduino Board or other controller board with I2C bus.
- Display Type: Negative white on Blue backlight.
- I2C Address: 0x38-0x3F (0x3F default)
- Supply voltage: 5V
- Interface: I2C to 4bits LCD data and control lines.
- Contrast Adjustment: built-in Potentiometer.
- Backlight Control: Firmware or jumper wire.
- Board Size: 80x36 mm.

Setting Up:

Hitachi's HD44780 based character LCD are very cheap and widely available, and is an essential part for any project that displays information. Using the LCD piggy-back board, desired data can be displayed on the LCD through the I2C bus. In principle, such backpacks are built around PCF8574 (from NXP) which is a general purpose bidirectional 8 bit I/O port expander that uses the I2C protocol. The PCF8574 is a silicon CMOS circuit provides general purpose remote I/O expansion (an 8-bit quasi-bidirectional) for most microcontroller families via the two-line bidirectional bus (I2C-bus). Note that most piggy-back modules are centered around PCF8574T (SO16 package of PCF8574 in DIP16 package) with a default slave address of 0x27. If your piggy-back board holds a PCF8574AT chip, then the default slave address will change to 0x3F. In short, if the piggy-back board is based on PCF8574T and the address connections (A0-A1-A2) are not bridged with solder it will have the slave address 0x27.



Address selection pads in the I2C-to-LCD piggy-back board.

Table 5. PCF8574A address map

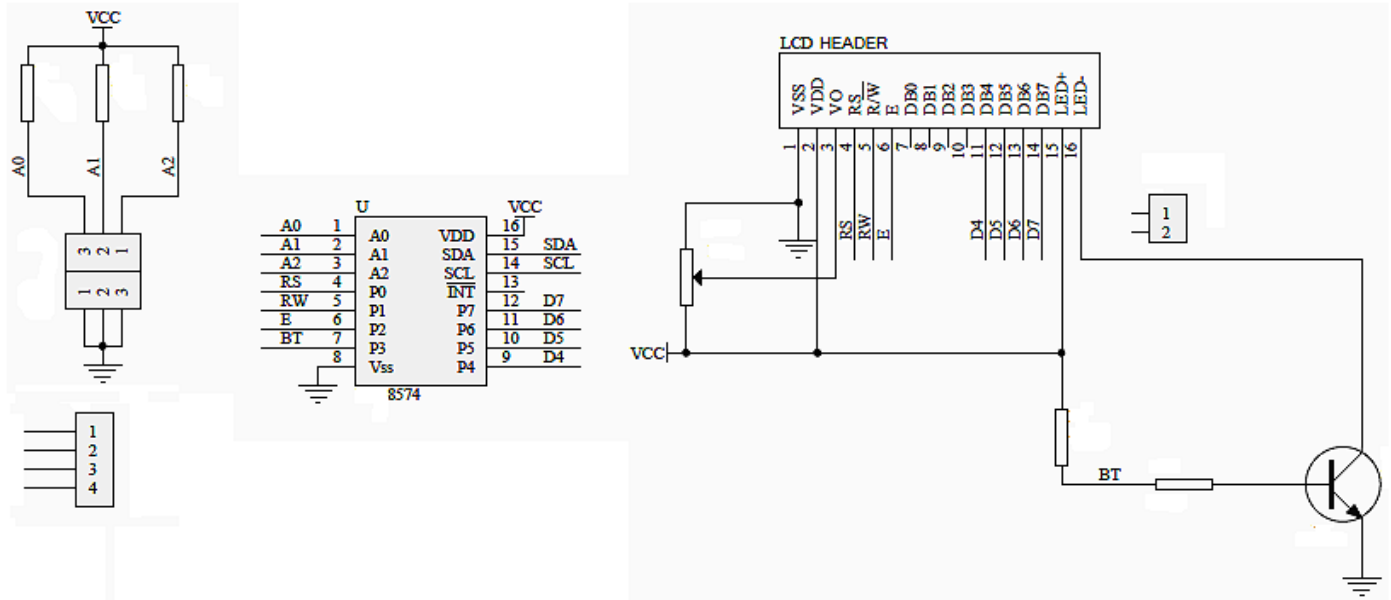
Pin connectivity			Address of PCF8574A								Address byte value		7-bit hexadecimal address without R/W
A2	A1	A0	A6	A5	A4	A3	A2	A1	A0	R/W	Write	Read	
V _{SS}	V _{SS}	V _{SS}	0	1	1	1	0	0	0	-	70h	71h	38h
V _{SS}	V _{SS}	V _{DD}	0	1	1	1	0	0	1	-	72h	73h	39h
V _{SS}	V _{DD}	V _{SS}	0	1	1	1	0	1	0	-	74h	75h	3Ah
V _{SS}	V _{DD}	V _{DD}	0	1	1	1	0	1	1	-	76h	77h	3Bh
V _{DD}	V _{SS}	V _{SS}	0	1	1	1	1	0	0	-	78h	79h	3Ch
V _{DD}	V _{SS}	V _{DD}	0	1	1	1	1	0	1	-	7Ah	7Bh	3Dh
V _{DD}	V _{DD}	V _{SS}	0	1	1	1	1	1	0	-	7Ch	7Dh	3Eh
V _{DD}	V _{DD}	V _{DD}	0	1	1	1	1	1	1	-	7Eh	7Fh	3Fh

Address Setting of PCD8574A (extract from PCF8574A data specs).

Note: When the pad A0~A2 is open, the pin is pull up to VDD. When the pin is solder shorted, it is pull down to VSS.

The default setting of this module is A0~A2 all open, so is pull up to VDD. The address is 3Fh in this case.

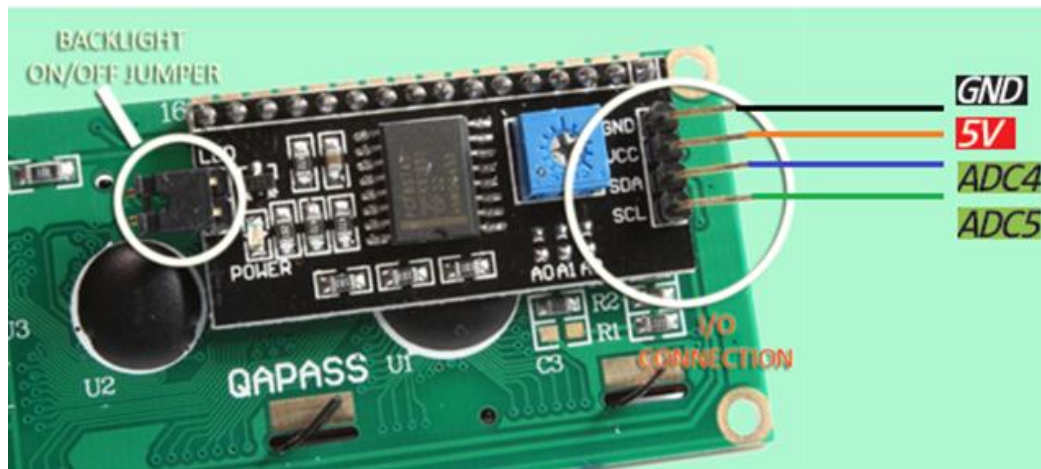
Reference circuit diagram of an Arduino-compatible LCD backpack is shown below. What follows next is information on how to use one of these inexpensive backpacks to interface with a microcontroller in ways it was exactly intended.



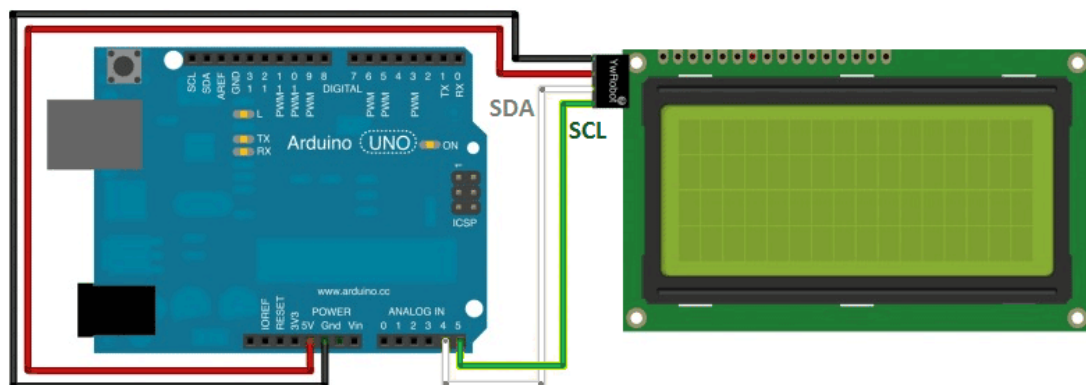
Reference circuit diagram of the I2C-to-LCD piggy-back board.

I2C LCD Display.

At first you need to solder the I2C-to-LCD piggy-back board to the 16-pins LCD module. Ensure that the I2C-to-LCD piggy-back board pins are straight and fit in the LCD module, then solder in the first pin while keeping the I2C-to-LCD piggy-back board in the same plane with the LCD module. Once you have finished the soldering work, get four jumper wires and connect the LCD module to your Arduino as per the instruction given below.



LCD display to Arduino wiring.



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